

This book provides a clear and concise introduction to the fundamental principles of biophysics and the working concepts of modern analytical instruments used in life sciences. It explores the physical laws governing biological systems, including molecular interactions, energy transformations, and structural organization, while also explaining the principles, applications, and significance of key instruments such as spectroscopy, chromatography, microscopy, and electrophoresis. Designed for students and researchers, this text bridges theory and practical application, offering a simplified yet comprehensive understanding of how biophysical techniques are used to analyze biological materials and solve real-world scientific problems.



**Dr. K. Mohan** is an accomplished academician and researcher, serving as Assistant Professor in the PG and Research Department of Zoology at Sri Vasavi College, Erode. He has over seven years of teaching and ten years of research experience. He was awarded DST-SERB National Post-Doctoral Fellowship and conducted research at Annamalai University.

# Book of Principles of Biophysics and Instrumentation

Mohan K.  
Kalaimani S.  
Revathi N.



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# **THE BOOK OF PRINCIPLES OF BIOPHYSICS AND INSTRUMENTATION**

First Edition-2026

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## FOREWORD

The book adheres to the scope and sequence of most one major course nationwide. also strive to make principles of Biophysics and Instrumentation, as a discipline, interesting and accessible to students. in addition to a comprehensive coverage of core concepts and foundational research, incorporated features that draw learners into the discipline in meaningful ways.

This book provides a fundamental concept of Principles of Biophysics and Instrumentation with a scope that is manageable for instructors and students alike with this; it is my pleasure to write a brief foreword to this wonderful preparatory book on “The Book of Principles of Biophysics and Instrumentation”.

I am appreciating the authors to chosen this book chapter on basic preparatory knowledge of Principles of Biophysics and Instrumentation. on the aspect of Biophysics and Instrumentation include number of books are available in the form of hand book, question bank for competitive examinations etc. except from this for the student community limited numbers of this like book are available in UG or PG level lack of syllabus covered text books are available, this book might be useful to basic and advanced learner of Biophysics and Instrumentation. written of this contain simple language and easy to understand a basic learner and valuable guide to the intellectual.

This book can help teachers develop a shared vision and understanding of basic concepts of Biophysics and Instrumentation and techniques of biology and its flexible uses across subjects and grade levels.

This book has been created with several goals in mind, accessibility, customization, and student engagement all while encouraging students toward high levels of academic scholarship. instructors and students alike will find that a strong introduction to Biophysics and Instrumentation in an accessible format.



Dr. S. UMAMAHESWARI  
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## PREFACE

A Students who has just entered the portals of colleges finds it difficult to understand the subject taught to him/her this difficulty is mainly due to understand standard references. While preparing this **“THE BOOK OF PRINCIPLES OF BIOPHYSICS AND INSTRUMENTATION”**. has been prepared to serve the special purpose of college students.

This book explains the Principles of Biophysics and Instrumentation in a simplified manner. This Preparatory book can be used for UG and PG Zoology students, who are looking for a quick revision before the Examination.

This book support to the zoology students to get confidence and also reference just before attending University Examination and other Competitive Examination.

This book has been written to Students and preparatory the field of Life sciences, no such book which cover suitable original picture with needed explanation of Principles of Biophysics and Instrumentation techniques and Practice.

This book is useful for the preparation of state Universities and College students integrated Certificate and Diploma courses field of Biophysics and Instrumentation.

**Dr. K. Mohan., M.Sc., M.Phil., Ph.D.,**

## ACKNOWLEDGEMENT

Words find no bounds to express my sincere thanks to all who made this Book to come true. my sincere thanks to **Dr.G.MALARVIZHI**, M.A., MPhil., Ph.D Principal (Ic), Sri Vasavi college, Erode, Erode D.t, for extending the facilities to undergo this Book work.

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My express my deep sense of feeling of my affectionate father **Mr.K.KANNAN**, my mother **Mrs.K.PUSHPA**, and my sister **Mrs.K.THANGAMANI**, to whom i owe everything for their dedicated efforts and loving encouragement to complete this book successfully.

Last but not least i wish to extend my sincere thanks to all my friends and relatives for their co-operation and support in this book to make success.

**Dr. K. Mohan., M.Sc., M.Phil., Ph.D.,**

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## **UNIT-I: BIOMOLECULAR INTERACTION**

### **BIOPHYSICS: INTRODUCTION**

Biophysics is an interdisciplinary science that applies the principles and methods of physics to understand biological systems. It bridges the gap between biology and physics, providing quantitative and theoretical insights into the mechanisms of life at molecular, cellular, and organismal levels.

Biophysics studies the structure, dynamics, and function of biological molecules (like DNA, proteins, lipids), as well as complex systems such as cells, tissues, and organs. Techniques from physics, including spectroscopy, electrophysiology, X-ray crystallography, and nuclear magnetic resonance (NMR), are widely used to analyze biological structures and processes.

The field encompasses various specialized areas such as:

- Molecular Biophysics – Structure and interactions of biomolecules.
- Cellular Biophysics – Membrane potentials, ion channels, and signaling.
- Medical Biophysics – Applications in imaging (MRI, CT), radiation therapy, and diagnostics.
- Computational Biophysics – Simulation and modeling of biological systems.

#### **Introduction to Biomolecular Interaction**

Biomolecular interactions are fundamental to all biological processes. They refer to the physical contacts and associations between molecules—such as proteins, nucleic acids, lipids, and carbohydrates—within a cell or an organism. These interactions are governed by non-covalent forces like hydrogen bonds, van der Waals forces, ionic bonds, and hydrophobic interactions.

#### **Importance**

Biomolecular interactions are essential for:

- Enzyme-substrate binding in catalysis
- Antigen-antibody recognition in immune response
- Receptor-ligand binding in signal transduction
- Protein-DNA interactions in gene regulation
- Protein-protein interactions in structural assemblies and signaling pathways

#### **Types of Biomolecular Interactions**

1. Protein-protein interactions – critical for complex formation and cellular signaling.
2. Protein-DNA/RNA interactions – crucial in replication, transcription, and translation.
3. Enzyme-substrate interactions – necessary for metabolic reactions.
4. Receptor-ligand interactions – central in hormonal and neurotransmitter signaling.
5. Antigen-antibody interactions – key in immune defense.

#### **Nature of Forces Involved**

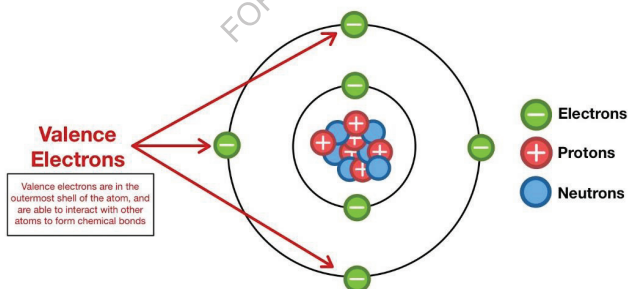
- Hydrogen bonding
- Electrostatic (ionic) interactions
- Hydrophobic interactions
- van der Waals forces

## VALENCE OF CARBON

The term "valence" (or valency) refers to the combining capacity of an element, especially how many chemical bonds an atom of the element can form with other atoms. It's a fundamental concept in chemistry that helps explain how atoms join together to form molecules.

Key Points on Valence:

1. Definition:
  - Valence is the number of electrons an atom uses to bond with other atoms.
  - It often corresponds to the number of electrons in the outermost shell (valence electrons).
2. In Terms of the Periodic Table:
  - Group 1 elements (like Na, K) have valence 1.
  - Group 2 elements (like Mg, Ca) have valence 2.
  - Group 17 (halogens like Cl, Br) have valence 1 (they need one electron).
  - Group 18 (noble gases like He, Ne) have valence 0 (they are inert).
3. Examples:
  - Hydrogen (H): valence = 1 → forms one bond (e.g.,  $\text{H}_2\text{O}$ )
  - Oxygen (O): valence = 2 → forms two bonds (e.g.,  $\text{H}_2\text{O}$ )
  - Carbon (C): valence = 4 → forms four bonds (e.g.,  $\text{CH}_4$ )
  - Nitrogen (N): valence = 3 → forms three bonds (e.g.,  $\text{NH}_3$ )
4. Types of Valency:
  - Primary valency – relates to oxidation state.
  - Variable valency – some elements (like Fe, Cu) show more than one valency.
    - Example:  $\text{Fe}^{2+}$  (valency 2),  $\text{Fe}^{3+}$  (valency 3)
5. Modern Concept:
  - In modern chemistry, valence is closely related to the number of unpaired electrons in the valence shell of an atom that can form bonds.



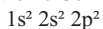
### Atomic Structure Highlighting Valence Electrons and Their Role in Chemical Bonding

## CARBON VALENCY

Electronic Configuration of Carbon

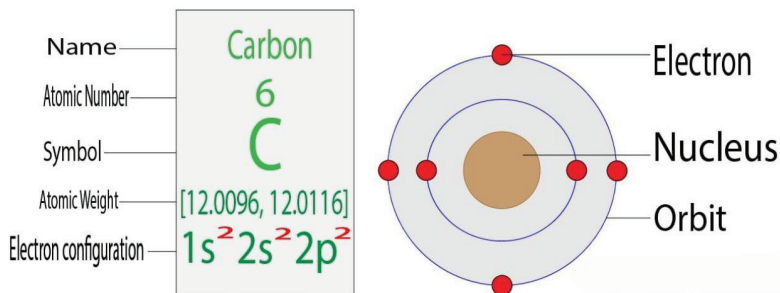
- Atomic Number: 6

- Electronic Configuration:



→ Total valence electrons = 4 (2 in 2s and 2 in 2p)

So, Carbon has 4 valence electrons in the outermost shell (second shell).



### Carbon Electron Configuration

Carbon has valency 4, meaning:

- It needs 4 more electrons to complete its octet (achieve 8 electrons in outer shell).
- It can form four covalent bonds with other atoms to achieve this.

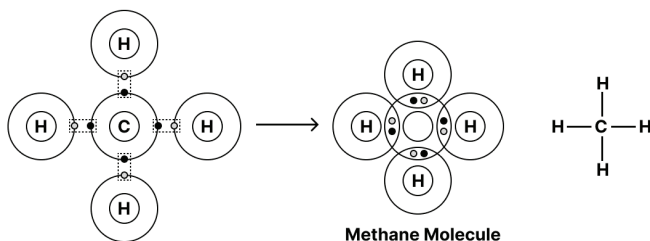
Carbon achieves stability by sharing its valence electrons — it forms covalent bonds, not ionic.

### TYPES OF BONDING BY CARBON

Carbon forms covalent bonds by sharing its electrons. Depending on how many pairs of electrons are shared, carbon can form:

#### a) Single Bond (Sigma bond, $\sigma$ )

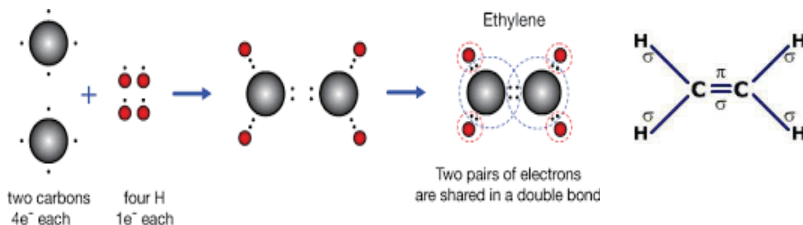
- Carbon shares one pair of electrons with another atom.
- Example: Methane ( $\text{CH}_4$ )
  - Each hydrogen shares 1 electron with carbon → 4 single bonds.



### Single Bond: Methane

b) Double Bond (1 Sigma + 1 Pi bond,  $\sigma + \pi$ )

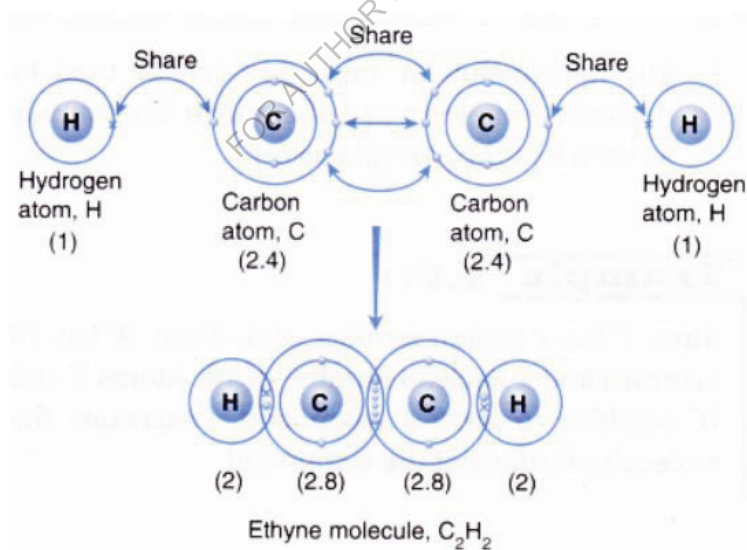
- Carbon shares two pairs of electrons.
- One  $\sigma$ -bond and one  $\pi$ -bond.
- Example: Ethene ( $\text{C}_2\text{H}_4$ )



**Double Bond: Ethene**

c) Triple Bond (1 Sigma + 2 Pi bonds,  $\sigma + 2\pi$ )

- Carbon shares three pairs of electrons.
- In the ethyne molecule,  $\text{C}_2\text{H}_2$ , a triple covalent bond is formed between the two carbon atoms and two single covalent bonds are formed between the carbon and hydrogen atoms.
- One  $\sigma$ -bond and two  $\pi$ -bonds.
- Example: Ethyne ( $\text{C}_2\text{H}_2$ )



**Formation of an Ethyne Molecule**

## POLAR AND NON-POLAR MOLECULES

### Introduction:

Molecules can be classified as polar or non-polar based on the distribution of electrons and the shape of the molecule, which affects their electrical dipole moment. This classification influences the physical properties, solubility, and intermolecular interactions of substances.

### 1. Polar Molecules:

#### Definition:

A polar molecule is a molecule in which the centers of positive and negative charges do not coincide, resulting in a net dipole moment.

#### Conditions for Polarity:

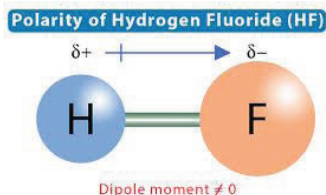
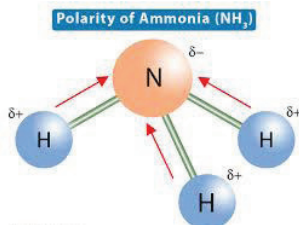
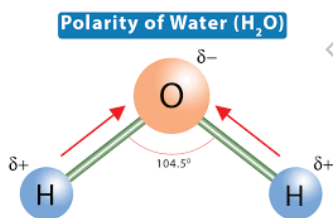
- Presence of polar bonds (difference in electronegativity between atoms).
- Asymmetrical shape of the molecule causing dipoles not to cancel out.

#### Characteristics:

- Unequal sharing of electrons.
- Have a dipole moment.
- Soluble in polar solvents (like water).
- Exhibit strong intermolecular forces (e.g., hydrogen bonding, dipole-dipole interaction).

#### Examples:

- Water ( $\text{H}_2\text{O}$ ) – bent shape and electronegativity difference between H and O.
- Ammonia ( $\text{NH}_3$ ) – trigonal pyramidal shape.
- Hydrogen fluoride ( $\text{HF}$ ) – large electronegativity difference between H and F.



### Polarity of Water, Ammonia and Hydrogen Fluoride

## 2. Non-Polar Molecules:

### Definition:

A non-polar molecule is one in which the electrons are shared equally between atoms, or the molecular shape causes bond dipoles to cancel each other out.

### Conditions for non-polarity:

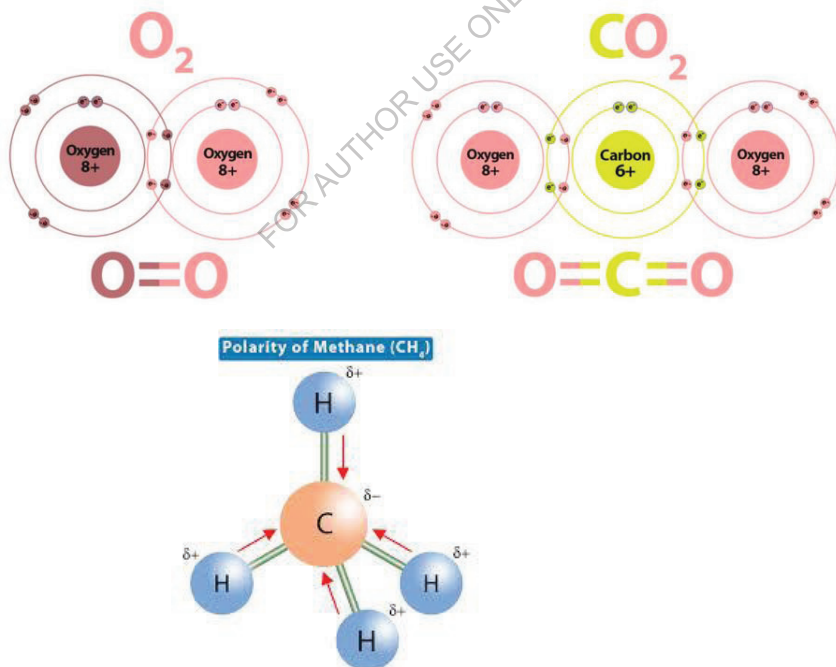
- Either the bonds are non-polar (same atoms or similar electronegativity).
- Or polar bonds are symmetrically arranged, resulting in cancellation of dipole moments.

### Characteristics:

- Equal sharing of electrons.
- No net dipole moment.
- Insoluble in polar solvents, but soluble in non-polar solvents (like benzene).
- Weaker intermolecular forces (e.g., London dispersion forces).

### Examples:

- Oxygen ( $O_2$ ) – identical atoms with equal sharing.
- Carbon dioxide ( $CO_2$ ) – linear geometry, dipoles cancel out.
- Methane ( $CH_4$ ) – symmetrical tetrahedral geometry.



**Non-Polar Molecules: Oxygen, Carbon di Oxide and Methane**

## COVALENT, IONIC AND CO-ORDINATE BONDS

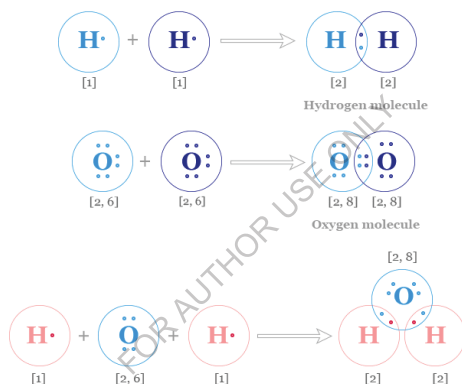
### 1. COVALENT BOND

#### Definition:

A covalent bond is formed by the mutual sharing of one or more pairs of electrons between two atoms, typically non-metals, to achieve stable electronic configuration (octet/duplet).

#### Characteristics:

- Directionality: Covalent bonds are directional.
- Formation: Occurs between atoms with similar electronegativities.
- Bond Strength: Stronger than ionic bonds under biological conditions.
- Examples:
  - $\text{H}_2$  (Hydrogen molecule)
  - $\text{O}_2$  (Oxygen molecule)
  - $\text{H}_2\text{O}$  (Water molecule)



#### Examples of Covalent Bonds

#### Types:

- Single bond: One pair of electrons shared (e.g.,  $\text{H}-\text{H}$ )
- Double bond: Two pairs of electrons shared (e.g.,  $\text{O}=\text{O}$ )
- Triple bond: Three pairs of electrons shared (e.g.,  $\text{N}\equiv\text{N}$ )



#### Types of Bonds

#### Biological Relevance:

- Found in DNA (phosphodiester bonds), proteins (peptide bonds).
- Enzyme-substrate interactions often involve covalent bonding.

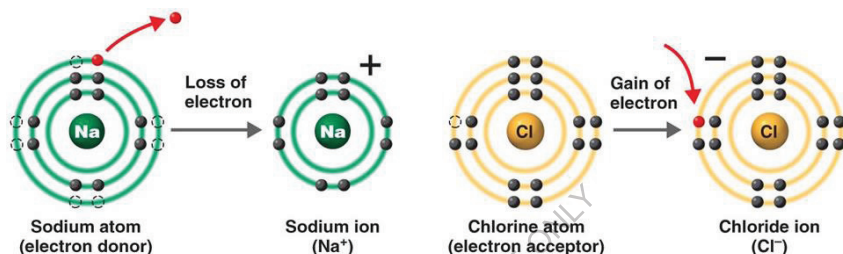
## 2. IONIC BOND or ELECTROSTATIC BOND

Definition:

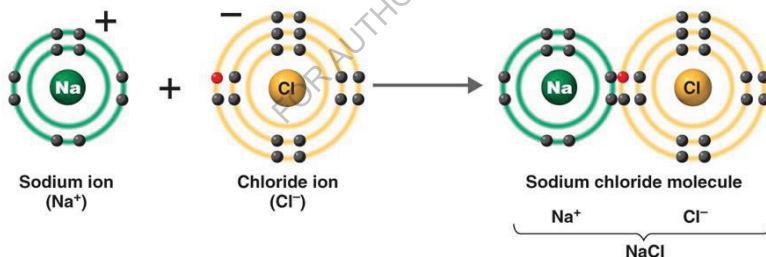
An ionic bond is formed by the complete transfer of electrons from a metal (electron donor) to a non-metal (electron acceptor), resulting in oppositely charged ions that attract each other.

Characteristics:

- Non-directional: Ionic bonds act equally in all directions.
- Formation: Between atoms with large electronegativity difference.
- Strength: Strong in solid state, but weakens in aqueous solution due to hydration.
- Examples:
  - NaCl (Sodium chloride)
  - KBr (Potassium bromide)



**(a)** A sodium atom (Na) loses one electron to an electron acceptor and forms a sodium ion ( $\text{Na}^+$ ). A chlorine atom (Cl) accepts one electron from an electron donor to become a chloride ion ( $\text{Cl}^-$ ).



**(b)** The sodium and chloride ions are attracted because of their opposite charges and are held together by an ionic bond to form a molecule of sodium chloride.

Biological Relevance:

- Found in salt bridges in proteins.
- Electrolytes (e.g.,  $\text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Ca}^{2+}$ ) crucial in nerve impulse transmission and muscle contraction.

## 3. CO-ORDINATE (DATIVE) COVALENT BOND

Definition:

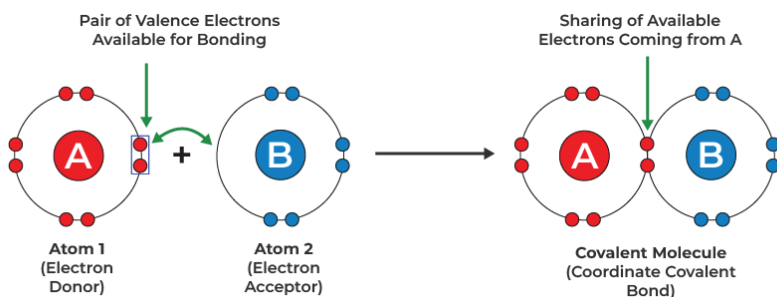
A coordinate bond is a type of covalent bond in which both electrons are donated by one atom, called the donor, to another atom (acceptor) which has a vacant orbital.

Characteristics:

- Represented as  $\rightarrow$  (e.g.,  $\text{NH}_3 \rightarrow \text{H}^+$ ).
- Donor atom must have a lone pair of electrons.
- Often found in complex ions, metal-ligand bonds, and enzyme-cofactor complexes.

Examples:

- $\text{NH}_4^+$  (Ammonium ion)
- $\text{H}_3\text{O}^+$  (Hydronium ion)
- $[\text{Cu}(\text{NH}_3)_4]^{2+}$  (Tetraammine copper complex)



### Coordinate Covalent Bond

Biological Relevance:

- Central to metal ion complexes in enzymes (e.g.,  $\text{Fe}^{2+}$  in hemoglobin).
- Chelation therapy relies on coordinate bonding to remove toxic metals.

## HYDROGEN BONDING-WEAK INTERACTIONS

Definition

Hydrogen bonding is a type of weak electrostatic attraction that occurs between a hydrogen atom covalently bonded to a highly electronegative atom (such as O, N, or F) and another electronegative atom with a lone pair of electrons.

Nature of Interaction

- Non-covalent, directional, and partially electrostatic in nature.
- Occurs intermolecularly (between molecules) or intramolecularly (within the same molecule).
- Stronger than van der Waals forces but weaker than covalent or ionic bonds.

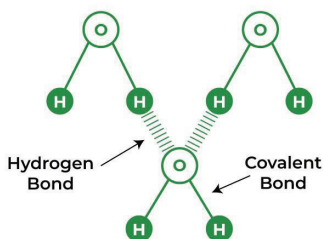
Conditions for Hydrogen Bonding

1. A hydrogen atom bonded to a strong electronegative atom (O, N, F).
2. Presence of a lone pair of electrons on a nearby electronegative atom.

### Types of Hydrogen Bonds

#### 1. Intermolecular Hydrogen Bond

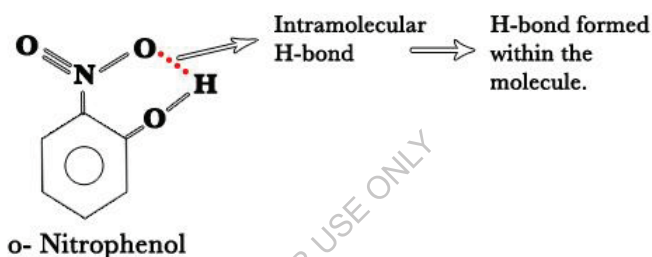
Intermolecular hydrogen bonding occurs between two separate molecules. Common examples include hydrogen bonding between water molecules ( $\text{H}_2\text{O}$ ), between ammonia molecules ( $\text{NH}_3$ ), and between water and ammonia.



Hydrogen Bonding in Water

## 2. Intramolecular Hydrogen Bond

Intramolecular hydrogen bonds occur within a single molecule, forming a ring-like structure. Examples include o-nitrophenol and salicylaldehyde.



### Intramolecular Hydrogen Bond

#### Strength and Energy

- Bond energy: 2–10 kcal/mol (8–40 kJ/mol)
- Distance: 1.5 to 2.5 Å
- Weaker than covalent bonds (typically 100–200 kcal/mol)

#### Biological Importance

1. Stabilizes secondary and tertiary structures of proteins ( $\alpha$ -helix,  $\beta$ -sheets).
2. Base pairing in DNA (A=T via 2 H-bonds; G=C via 3 H-bonds).
3. Important in enzyme-substrate binding.
4. Contributes to the high boiling point of water and its solvent properties.

#### Examples

| System   | Hydrogen Bonding        |
|----------|-------------------------|
| Water    | H–O...H                 |
| DNA      | N–H...O and N–H...N     |
| Alcohols | O–H...O                 |
| Proteins | N–H...O (peptide bonds) |

## ESTER LINKAGE

### Definition:

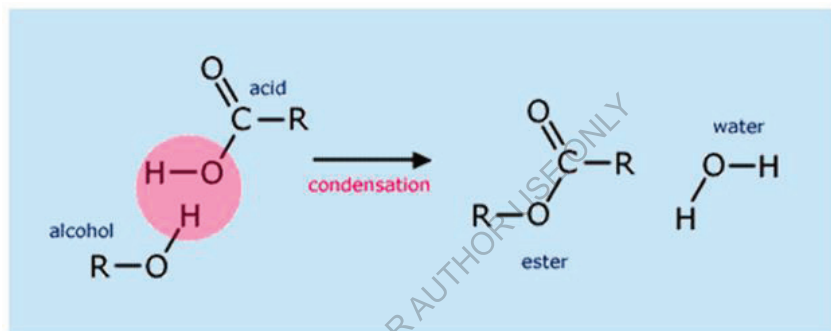
An ester linkage is a covalent bond formed between a carboxylic acid (-COOH) and an alcohol (-OH) group, resulting in the formation of an ester (-COOR) and the release of a molecule of water (H<sub>2</sub>O).



Carboxylic acid + Alcohol  $\rightarrow$  Ester + Water

The bond formed between the carbon atom of -COOH and the oxygen atom of -OH is called the ester bond or ester linkage.

This type of reaction is called a condensation reaction or esterification.



### Biological Importance:

#### 1. Lipids (Triglycerides):

- Ester linkages form between glycerol and fatty acids.
- Each molecule of glycerol forms three ester linkages with three fatty acids.

#### 2. Phospholipids:

- Found in biological membranes.
- Contain ester bonds between glycerol backbone and fatty acid chains.

#### 3. Waxes:

- Long-chain fatty acids + long-chain alcohols = esters (wax esters).

#### 4. Aspirin and Esters in Medicine:

- Ester functional groups are present in drugs like aspirin (acetylsalicylic acid).

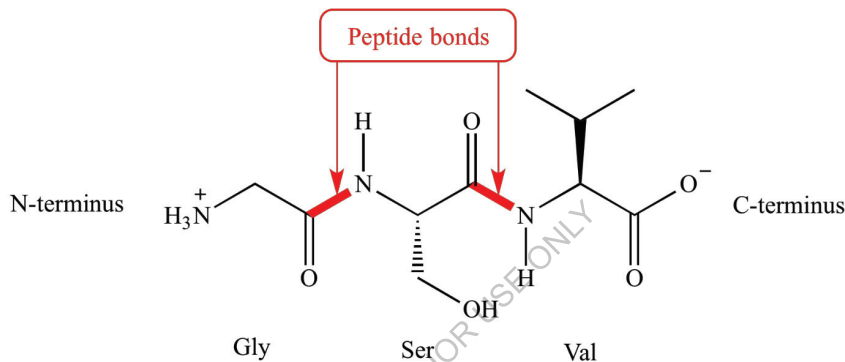
## PEPTIDE AND DISULPHIDE BONDS

### Peptide Bond (Amide Bond)

#### Definition:

A peptide bond is a covalent bond formed between the carboxyl group ( $-\text{COOH}$ ) of one amino acid and the amino group ( $-\text{NH}_2$ ) of another, with the release of a water molecule.

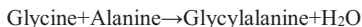
Reaction (Condensation reaction):



#### Characteristics:

- Rigid and planar due to partial double bond character (resonance).
- Usually in trans configuration.
- Not easily broken (requires enzymatic hydrolysis or strong acid/base).
- Backbone of proteins and peptides.

#### Example:



#### Importance:

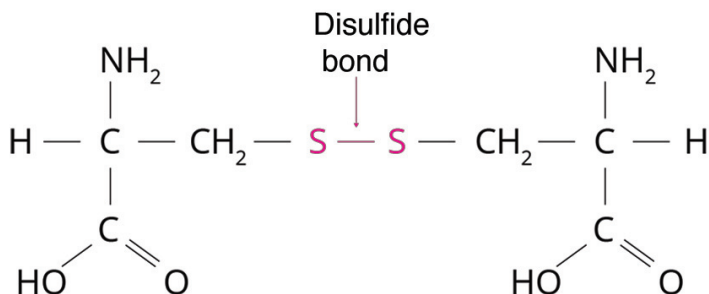
- Forms the primary structure of proteins.
- Determines the sequence of amino acids.

## 2. Disulphide Bond (Disulfide Bridge)

#### Definition:

A disulphide bond is a covalent bond formed between two thiol groups ( $-\text{SH}$ ) of cysteine residues in a protein.

**Reaction:**



Characteristics:

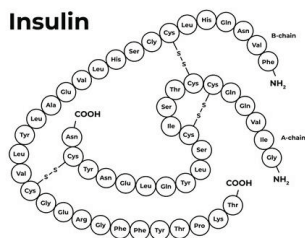
- Forms via oxidation of sulfhydryl (-SH) groups.
- Found in extracellular or secreted proteins.
- Can be intra-chain (within same polypeptide) or inter-chain (between two polypeptides).

**Function:**

- Stabilizes the tertiary and quaternary structures of proteins.
- Helps maintain protein conformation under stress conditions.

**Examples:**

- Insulin: Contains inter- and intra-chain disulphide bonds.
- Keratin: Structural protein in hair, rich in disulphide bridges.



## VANDER-WAALS FORCES

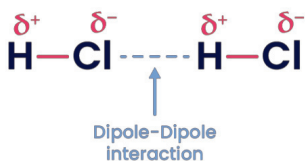
**Definition:**

Van der Waals forces are weak, non-covalent intermolecular forces that arise between molecules or atoms due to transient or permanent electric dipoles. They are essential in explaining the physical properties of gases, liquids, and solids (especially molecular crystals and biological molecules).

## Types of Van der Waals Forces:

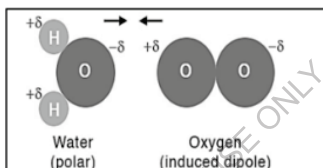
### 1. Dipole–Dipole Interactions:

- Occur between polar molecules.
- Positive end of one molecule attracts the negative end of another.
- Example: Interaction between HCl molecules.



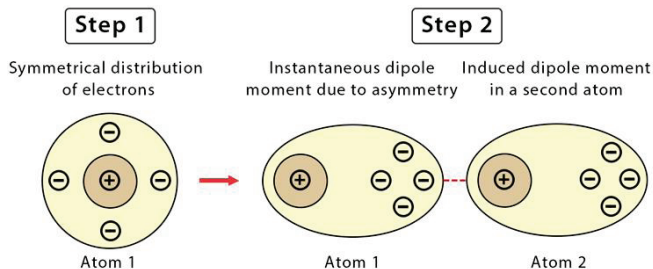
### 2. Dipole–Induced Dipole Interactions:

- A polar molecule induces a dipole in a non-polar molecule by disturbing its electron cloud.
- Example: Oxygen ( $O_2$ ) dissolving in water.



### 3. London Dispersion Forces (Instantaneous Dipole–Induced Dipole):

- Present in all molecules, especially non-polar ones.
- Arise due to temporary fluctuations in electron distribution.
- Strength increases with molecular size and surface area.
- Example: Interaction between noble gas atoms like Argon.



## London Dispersion Forces

### Characteristics:

- Weakest among all intermolecular forces (except for gravitational).
- Short-range forces – effective only at very close distances.
- Play a major role in biological systems (e.g., protein folding, DNA base stacking).
- Important in physical properties such as boiling and melting points, surface tension, and viscosity.

### Factors Affecting Van der Waals Forces:

1. Size of the molecule: Larger molecules have stronger dispersion forces.
2. Shape of the molecule: More surface area (linear shape) → stronger forces.
3. Polarity: Polar molecules have stronger dipole-related interactions.

## ISOMERISM

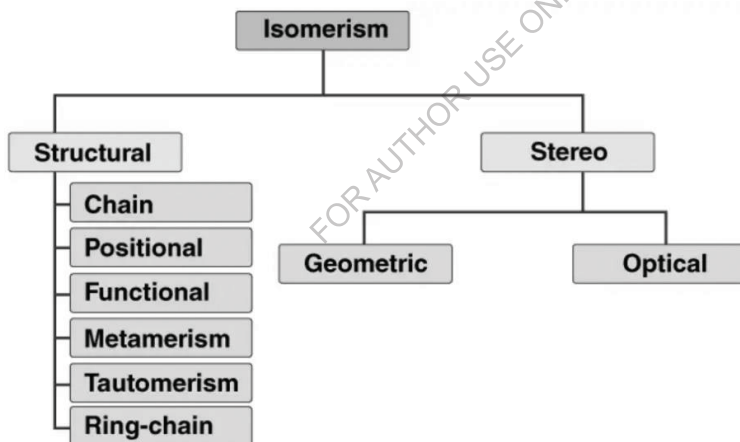
### Definition:

Chemical compounds that have identical chemical formulae but differ in properties and the arrangement of atoms in the molecule are called isomers. Therefore, the compounds that exhibit isomerism are known as isomers.

The word “isomer” is derived from the Greek words “isos” and “meros”, which mean “equal parts”. This term was coined by the Swedish chemist Jacob Berzelius in the year 1830.

### Types of Isomerism:

There are two primary types of isomerism, which can be further categorized into different subtypes. These primary types are Structural Isomerism and Stereoisomerism.

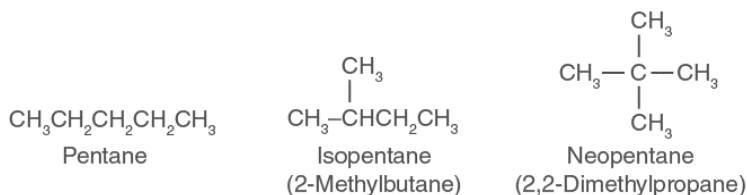


#### I. Structural Isomerism

Structural isomerism is commonly referred to as constitutional isomerism. The functional groups and the atoms in the molecules of these isomers are linked in different ways. Different structural isomers are assigned different IUPAC names since they may or may not contain the same functional group.

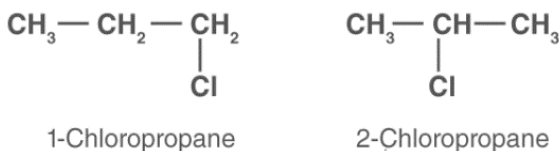
##### (i). Chain Isomerism

- It is also known as skeletal isomerism.
- The components of these isomers display differently branched structures.
- Commonly, chain isomers differ in the branching of carbon
- An example of chain isomerism can be observed in the compound  $C_5H_{12}$ , as illustrated below.



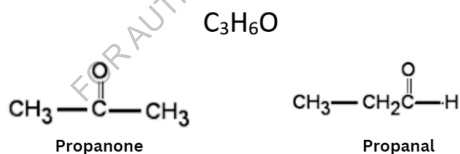
(ii). Position Isomerism

- The positions of the functional groups or substituent atoms are different in position isomers.
- Typically, this isomerism involves the attachment of the functional groups to different carbon atoms in the carbon chain.
- An example of this type of isomerism can be observed in the compounds having the formula  $\text{C}_3\text{H}_7\text{Cl}$ .



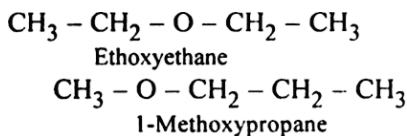
(iii). Functional Isomerism

- It is also known as functional group isomerism.
- As the name suggests, it refers to the compounds that have the same chemical formula but different functional groups attached to them.
- An example of functional isomerism can be observed in the compound  $\text{C}_3\text{H}_6\text{O}$ .



(iv). Metamerism

- This type of isomerism arises due to the presence of different alkyl chains on each side of the functional group.
- It is a rare type of isomerism and is generally limited to molecules that contain a divalent atom (such as sulphur or oxygen), surrounded by alkyl groups.
- Example:  $\text{C}_4\text{H}_{10}\text{O}$  can be represented as ethoxyethane ( $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ ) and methoxy-propane ( $\text{CH}_3\text{OC}_3\text{H}_7$ ).



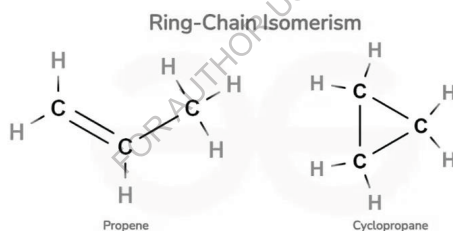
### (v). Tautomerism

- A tautomer of a compound refers to the isomer of the compound which only differs in the position of protons and electrons.
- Typically, the tautomers of a compound exist together in equilibrium and easily interchange.
- It occurs via an intramolecular proton transfer.
- An important example of this phenomenon is Keto-enol tautomerism.



### (vi) Ring-Chain Isomerism

- In ring-chain isomerism, one of the isomers has an open-chain structure whereas the other has a ring structure.
- They generally contain a different number of pi bonds.
- A great example of this type of isomerism can be observed in  $C_3H_6$ . Propene and cyclopropane are the resulting isomers, as illustrated below.

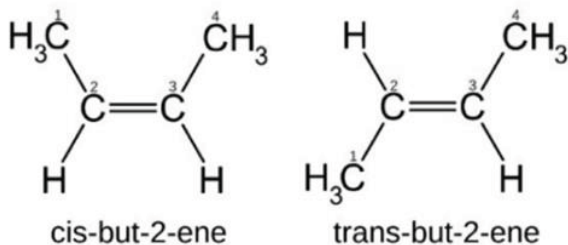


## II. Stereoisomerism

This type of isomerism arises in compounds having the same chemical formula but different orientations of the atoms belonging to the molecule in three-dimensional space. The compounds that exhibit stereoisomerism are often referred to as stereoisomers. This phenomenon can be further categorized into two subtypes.

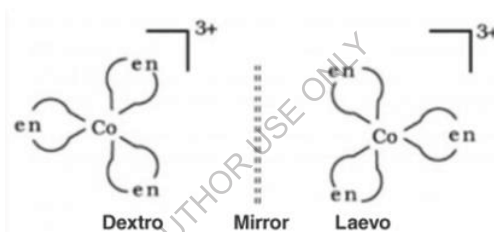
### (i). Geometric Isomerism

- It is popularly known as cis-trans isomerism.
- These isomers have different spatial arrangements of atoms in three-dimensional space.
- An illustration describing the geometric isomerism observed in the acyclic but-2-ene molecule is provided below.



(ii). Optical Isomerism

- Compounds that exhibit optical isomerism feature similar bonds but different spatial arrangements of atoms forming non-superimposable mirror images.
- These optical isomers are also known as enantiomers.
- Enantiomers differ from each other in their optical activities.
- Dextro enantiomers rotate the plane of polarized light to the right whereas laevo enantiomers rotate it to the left, as illustrated below.



## OPTICAL ACTIVITY

### Definition:

Optical activity is the ability of a compound to rotate the plane of plane-polarized light.

### Key Concepts:

#### 1. Chirality and Chiral Center:

- A carbon atom attached to four different groups is chiral.
- A molecule with at least one chiral center may show optical activity.
- Chiral molecules are non-superimposable on their mirror images.

#### 2. Enantiomers:

- Pair of non-superimposable mirror image isomers.
- Have identical physical properties (melting point, boiling point) except:
  - Rotate plane-polarized light in opposite directions.

- React differently with other chiral compounds (e.g., enzymes).

3. Dextrorotatory (+):

- Rotates light clockwise (to the right).

4. Levorotatory (−):

- Rotates light counter-clockwise (to the left).

5. Racemic Mixture (±):

- Equimolar mixture of both enantiomers.
- Optically inactive due to mutual cancellation of rotations.

6. Diastereomers:

- Stereoisomers that are not mirror images.
- Have different physical and chemical properties.

Optical Isomerism in Biochemistry:

- Amino acids (except glycine) are chiral and optically active.
- Natural amino acids are L-isomers.
- Sugars show D- and L- configurations based on the configuration of the chiral carbon farthest from the carbonyl group.

Measurement of Optical Activity

Optical activity, the ability of a substance to rotate the plane of polarized light, is measured using a polarimeter. This instrument passes plane-polarized light through a sample and measures the angle of rotation. The measured angle is then used to determine the specific rotation, which is a characteristic property of chiral molecules.

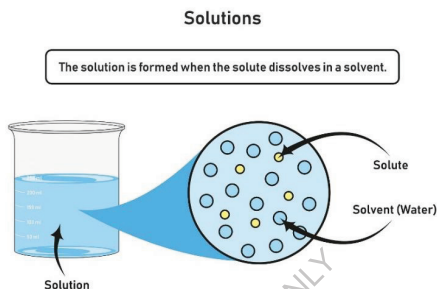
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## UNIT II: SOLUTIONS

### 1. Definition of a Solution:

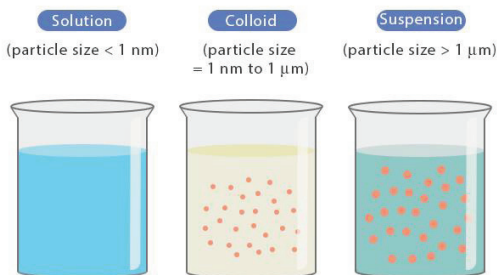
A solution is a homogeneous mixture of two or more substances. In biophysics, solutions are important for understanding how biological molecules behave in water and other solvents.

- Solvent: The substance in greater amount (usually water in biological systems).
- Solute: The substance that is dissolved.



### 2. Types of Solutions:

| Type of Solution   | Example in Biophysics               |
|--------------------|-------------------------------------|
| True Solution      | Glucose in water                    |
| Colloidal Solution | Proteins in plasma                  |
| Suspension         | Cells in blood (not true solutions) |



### **Types of Solutions**

### 3. Properties of Solutions:

#### a. Concentration Terms:

- Molarity (M) = moles of solute / liter of solution
- Molality (m) = moles of solute / kg of solvent
- Normality (N) = equivalents of solute / liter of solution
- Percent solutions: e.g., 0.9% NaCl = 0.9 g NaCl per 100 ml solution

b. Colligative Properties (depend on number, not type of particles):

- Osmotic Pressure – Crucial in cell osmoregulation
- Boiling Point Elevation
- Freezing Point Depression
- Vapour Pressure Lowering

4. Osmosis and Tonicity:

a. Osmosis:

Movement of water from low solute concentration to high solute concentration across a semipermeable membrane.

b. Tonicity Types:

| Type       | Effect on Cells                         |
|------------|-----------------------------------------|
| Isotonic   | No net water movement (e.g., 0.9% NaCl) |
| Hypotonic  | Water enters cell (swells or bursts)    |
| Hypertonic | Water leaves cell (shrinks/crenates)    |

5. Biological Importance of Solutions:

- Buffer Solutions: Maintain pH (e.g., phosphate buffer, bicarbonate buffer).
- Electrolyte Solutions: Conduct electrical impulses ( $\text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Cl}^-$ ).
- Drug Delivery Systems: Use of isotonic and pH-adjusted solutions.
- Dialysis: Removal of waste solutes via diffusion across a membrane.
- Enzyme Activity: Affected by ionic strength and solute concentration.

6. Examples in Biophysics Applications:

- Plasma: A colloidal solution of proteins (albumin, globulin).
- Intravenous Fluids: Must be isotonic to prevent hemolysis or shrinkage.
- Nernst Equation: Describes ion distribution across membranes.
- Henderson-Hasselbalch Equation: Describes buffer systems in blood.

## I. HYDROPHILES AND HYDROPHOBES

1. Hydrophiles

Hydrophiles are substances that are "water-loving" and can dissolve easily in water.

Characteristics:

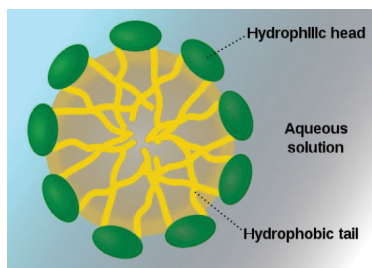
- Polar in nature.
- Form hydrogen bonds with water.
- Dissolve due to attraction between solute and water molecules.

Examples:

- Salts (e.g., NaCl)
- Sugars (e.g., glucose)
- Alcohols (e.g., ethanol)
- Amino acids

Applications:

- In biological systems, hydrophilic molecules (like proteins and carbohydrates) interact well with water.
- Used in making aqueous drugs and biomedical solutions.



## Difference Between Hydrophobic and Hydrophilic Molecules

### 2. Hydrophobes

Hydrophobes are substances that are "water-fearing" and do not dissolve in water.

Characteristics:

- Non-polar in nature.
- Cannot form hydrogen bonds with water.
- Tend to aggregate or separate from water (phase separation).

Examples:

- Oils (e.g., vegetable oil)
- Fats and waxes
- Hydrocarbons (e.g., benzene)
- Long-chain alkanes

Behavior:

- Hydrophobic substances often form micelles or bubbles in water.
- Important in cell membrane formation (phospholipid bilayer).

### 3. Hydrophilic and Hydrophobic Interactions

- Hydrophilic interactions promote solubility and dispersion in water.
- Hydrophobic interactions lead to aggregation (important in protein folding and membrane formation).

### 4. Biological Significance

- Proteins have both hydrophilic (surface-exposed) and hydrophobic (core) regions.
- Cell membranes are made of hydrophobic lipid tails and hydrophilic heads.
- Helps in drug delivery systems (liposomes, micelles).

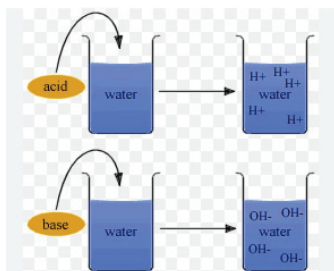
## II. ACID-BASE CONCEPT

### 1. Introduction

Acid-base theories explain the behavior of substances in aqueous and non-aqueous solutions. These concepts help in understanding chemical reactions such as neutralization, hydrolysis, and buffer action.

### 2. Classical Arrhenius Concept (1884)

- Acid: Produces  $H^+$  ions in aqueous solution.  
*Example:*  $HCl \rightarrow H^+ + Cl^-$
- Base: Produces  $OH^-$  ions in aqueous solution.  
*Example:*  $NaOH \rightarrow Na^+ + OH^-$



### Classical Arrhenius Concept

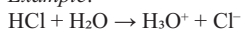
Limitations:

- Only applies to aqueous solutions.
- Does not explain acid-base reactions in non-aqueous media.
- Cannot explain ammonia acting as a base ( $\text{NH}_3$  has no  $\text{OH}^-$ ).

### 3. Brønsted–Lowry Concept (1923)

- Acid: Proton ( $\text{H}^+$ ) donor.
- Base: Proton ( $\text{H}^+$ ) acceptor.

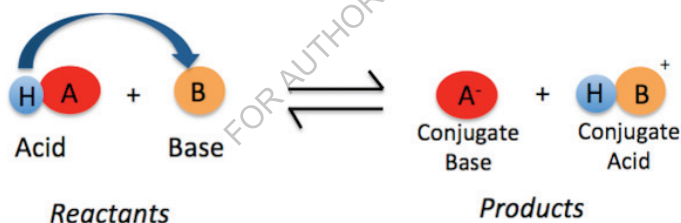
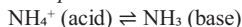
*Example:*



Here,  $\text{HCl}$  is an acid (donor),  $\text{H}_2\text{O}$  is a base (acceptor).

- Conjugate acid-base pairs: Acid and base differ by one proton.

*Example:*



### Brønsted-Lowry Acid-Base Reaction

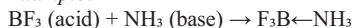
Advantages:

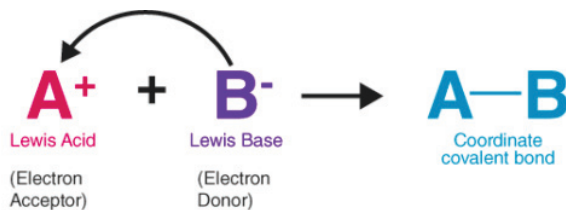
- Works in aqueous and non-aqueous media.
- Explains acid-base behavior of substances without  $\text{OH}^-$ .

### 4. Lewis Concept (1923)

- Acid: Electron pair acceptor.
- Base: Electron pair donor.

*Example:*





### Lewis Concept

Advantages:

- Broadest concept of acid-base.
- Includes metal ions and complex formation reactions.

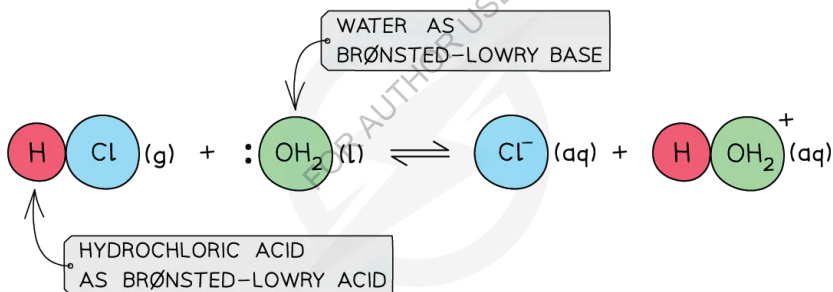
### 5. Amphoteric Substances

Definition: A substance that can act both as an acid and a base depending on the medium.

- Also known as amphoteric substances
- Can donate or accept protons ( $H^+$ )

Examples:

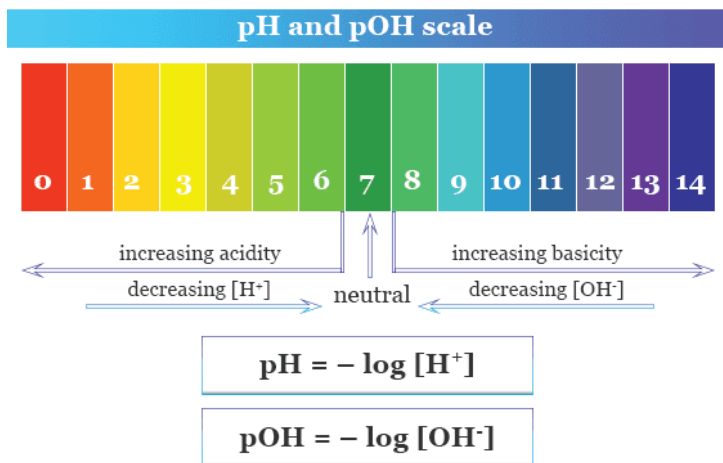
- Water ( $H_2O$ ):
  - Acts as acid with  $NH_3$ :  $H_2O \rightarrow OH^- + H^+$
  - Acts as base with  $HCl$ :  $H_2O + H^+ \rightarrow H_3O^+$
- Amino Acids: Have both  $-NH_2$  (base) and  $-COOH$  (acid) groups
- Hydrogen carbonate ( $HCO_3^-$ ), zwitterions, metal oxides like  $Al_2O_3$



### Amphoteric Substances

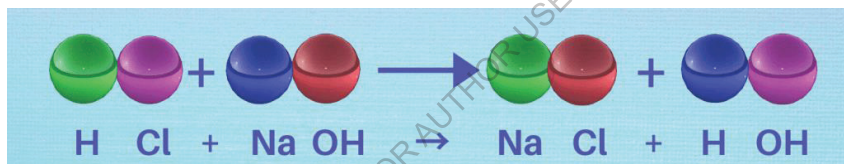
#### 6. pH and pOH

- pH:  $-\log[H^+]$
- pOH:  $-\log[OH^-]$
- Relation:  $pH + pOH = 14$  (at  $25^\circ C$ )



#### 7. Neutralization Reaction

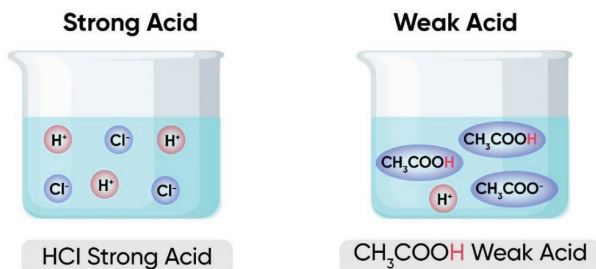
- Acid + Base  $\rightarrow$  Salt + Water  
*Example:*  $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$



**Neutralization Reaction**

#### 8. Acid and Base Strength

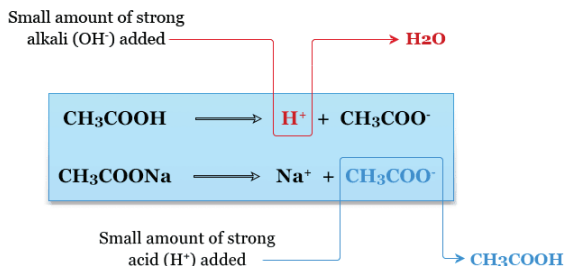
- Strong acids/bases: Completely ionized in solution.  
*Examples:*  $\text{HCl}$ ,  $\text{HNO}_3$ ,  $\text{NaOH}$ ,  $\text{KOH}$
- Weak acids/bases: Partially ionized.  
*Examples:*  $\text{CH}_3\text{COOH}$ ,  $\text{NH}_3$



## 9. Buffer Solutions

- Resist change in pH upon addition of small amounts of acid/base.
- Made of weak acid + salt or weak base + salt.

*Example:*  $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$



## 10. Applications

- Biology: Enzyme activity is pH-sensitive.
- Medicine: Blood pH buffering is crucial.
- Industry: Acid-base reactions used in titration, dye production, fertilizers.

## II. MOLARITY, MOLALITY AND NORMALITY

### 1. MOLARITY (M)

Definition: Number of moles of solute present in 1 litre (1000 mL) of solution.

Formula:

Molarity (M) = Moles of solute / Volume of solution in litres

Unit: mol/L or M

Example:

2 moles of NaCl dissolved in 1 litre of water  $\rightarrow$  2 M NaCl solution.

Applications:

- Titration calculations: Used to determine unknown concentrations using known volumes.
- Pharmaceutical preparations: Dosage of drugs is often based on molar concentration.
- Solution preparation in labs: Standard solutions in chemistry use molarity.
- Chemical kinetics: Reaction rates often depend on molar concentrations.
- Thermodynamics: Colligative properties (like boiling point elevation) often use molar concentration.

### 2. MOLALITY (m)

Definition: Number of moles of solute present in 1 kg (1000 g) of solvent.

Formula:

Molality (m) = Moles of solute / Mass of solvent in kg

Unit: mol/kg

Example:

2 moles of NaCl in 1 kg of water  $\rightarrow$  2 m NaCl solution.

Applications:

- Temperature-independent concentration: Molality doesn't change with temperature (unlike molarity), so it's useful in:
  - Boiling point elevation
  - Freezing point depression
  - Osmotic pressure studies

- Thermochemistry: Used in calculating enthalpy changes where mass-based measurements are involved.
- Cryoscopic and ebullioscopy constant experiments

### 3. NORMALITY (N)

Definition: Number of gram equivalents of solute per litre of solution.

Formula:

Normality (N) = Number of grams equivalents / Volume of solution in litres

Gram Equivalent = Molar mass / n-factor (valency/equivalency)

Unit: eq/L or N

Example:

- 1 M  $\text{H}_2\text{SO}_4$  (diprotic acid,  $n = 2$ )  $\rightarrow$  2 N solution

#### Applications

- Acid-base titrations: Easily used in titration calculations involving  $\text{H}^+$  or  $\text{OH}^-$  ions.
  - Example: 1N  $\text{H}_2\text{SO}_4$  provides 2 moles of  $\text{H}^+$ .
- Redox reactions: Useful for electron transfer equivalents.
  - Example:  $\text{KMnO}_4$  solutions in permanganate titrations.
- Precipitation reactions: Normality helps determine ion equivalents.
- Water hardness: Often measured using EDTA in terms of normality.

#### Comparison Table: Molarity vs Molality vs Normality

| Property        | Molarity (M)                                                          | Molality (m)                                                        | Normality (N)                                                                                   |
|-----------------|-----------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Definition      | Moles of solute per liter of solution                                 | Moles of solute per kilogram of solvent                             | Gram equivalent of solute per liter of solution                                                 |
| Formula         | $M = \frac{\text{moles of solute}}{\text{volume of solution (L)}}$    | $m = \frac{\text{moles of solute}}{\text{mass of solvent (kg)}}$    | $N = \frac{\text{equivalents of solute}}{\text{volume of solution (L)}}$                        |
| Units           | mol/L                                                                 | mol/kg                                                              | eq/L                                                                                            |
| Depends on      | Volume of solution                                                    | Mass of solvent                                                     | Volume of solution and equivalent weight                                                        |
| Effect of Temp. | Changes with temperature (volume changes)                             | Independent of temperature (mass remains constant)                  | Changes with temperature (volume changes)                                                       |
| Use in          | General chemistry, titrations                                         | Thermodynamic studies, boiling/freezing point                       | Acid-base & redox titrations (based on equivalent concept)                                      |
| Example         | 1 M $\text{NaCl} \rightarrow$ 1 mole $\text{NaCl}$ in 1 L of solution | 1 m $\text{NaCl} \rightarrow$ 1 mole $\text{NaCl}$ in 1 kg of water | 1 N $\text{H}_2\text{SO}_4 \rightarrow$ 1 eq. (49 g) $\text{H}_2\text{SO}_4$ in 1 L of solution |

## III. pH AND PKA VALUE

### 1. pH (Potential of Hydrogen)

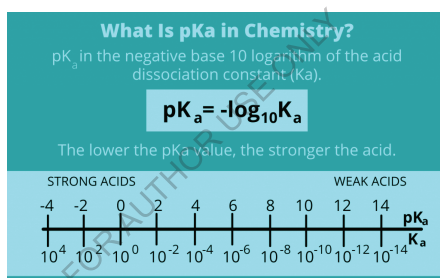
- Definition:
  - pH is a measure of the hydrogen ion concentration  $[\text{H}^+]$  in a solution.
- Formula:
  - $\text{pH} = -\log_{10}[\text{H}^+]$
- pH Scale:
  - Ranges from 0 to 14
  - $\text{pH} < 7$  = Acidic
  - $\text{pH} = 7$  = Neutral
  - $\text{pH} > 7$  = Basic (Alkaline)

- Examples:
  - Gastric juice: ~1.5–3.5
  - Pure water: 7.0
  - Blood: 7.35–7.45
- Importance:
  - Enzyme activity is pH-sensitive
  - pH affects protein structure and cell metabolism
  - Vital for buffer systems in biological fluids

## 2. pKa (Acid Dissociation Constant)

- Definition:
 

pKa is the negative logarithm of the acid dissociation constant (Ka) of a weak acid.
- Formula:
 
$$\text{pKa} = -\log_{10} K_a$$
- Relation with strength:
  - Lower pKa = Stronger acid
  - Higher pKa = Weaker acid
- Example:
  - Acetic acid ( $\text{CH}_3\text{COOH}$ ):  
 $K_a \approx 1.8 \times 10^{-5}$ ,  $\text{pKa} \approx 4.76$



## 3. Relation Between pH and pKa: Henderson-Hasselbalch Equation

Used to calculate the pH of buffer solutions:

$$\text{pH} = \text{pKa} + \log \left( \frac{[\text{A}^-]}{[\text{HA}]}\right)$$

Where:

- $[\text{A}^-]$  = concentration of conjugate base
- $[\text{HA}]$  = concentration of acid

## 4. Applications in Biology and Chemistry

- Buffers:
  - Maintain stable pH in cells and blood
  - Common buffer: Bicarbonate buffer system in blood
- Protein Chemistry:
  - Amino acid ionization depends on pKa
  - Affects protein folding, activity, and solubility
- Drug Design:
  - Drug absorption depends on pKa and pH of the environment

## Important Biological pKa Values

| Molecule                                     | pKa Value |
|----------------------------------------------|-----------|
| Acetic acid                                  | ~4.76     |
| Lactic acid                                  | ~3.86     |
| Phosphoric acid (1st)                        | ~2.15     |
| Phosphoric acid (2nd)                        | ~7.2      |
| Ammonium ion (NH <sub>4</sub> <sup>+</sup> ) | ~9.25     |
| Histidine (side chain)                       | ~6.0      |
| Carbonic acid                                | ~6.1      |

## IV. REDOX POTENTIAL (OXIDATION-REDUCTION POTENTIAL)

Definition:

Redox potential (also called oxidation-reduction potential or Eh) is a measure of the tendency of a chemical species to acquire electrons and thereby be reduced. It is measured in volts (V) or millivolts (mV).

Key Concepts:

| Term                 | Description                                                |
|----------------------|------------------------------------------------------------|
| Oxidation            | Loss of electrons (increase in oxidation state)            |
| Reduction            | Gain of electrons (decrease in oxidation state)            |
| Redox Reaction       | A reaction involving both oxidation and reduction          |
| Electron Donor       | Substance that loses electrons (gets oxidized)             |
| Electron Acceptor    | Substance that gains electrons (gets reduced)              |
| Redox Potential (Eh) | Indicates how readily a substance gains or loses electrons |

Measurement:

- Measured using a platinum electrode in comparison with a standard hydrogen electrode (SHE).
- Expressed in millivolts (mV) or volts (V).

Standard Redox Potential (E°):

- Measured under standard conditions:
  - 1 M concentration
  - 25°C (298 K)
  - 1 atm pressure
- E° > 0: Strong oxidizing agent
- E° < 0: Strong reducing agent

Examples of Redox Potentials:

| Half-Reaction                                                                     | E° (V) |
|-----------------------------------------------------------------------------------|--------|
| $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$          | +1.23  |
| $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$                          | +0.77  |
| $\text{NAD}^+ + \text{H}^+ + 2\text{e}^- \rightarrow \text{NADH}$                 | -0.32  |
| $\text{H}^+ + \text{e}^- \rightarrow \frac{1}{2} \text{H}_2$ (standard electrode) | 0.00   |

**Biological Importance:**

1. Cellular Respiration:
  - In mitochondria, redox reactions drive ATP synthesis (electron transport chain).
  - Oxygen is the terminal electron acceptor with high positive redox potential.

2. Photosynthesis:
  - In chloroplasts, redox reactions occur in photosystems I and II.
  - Water is oxidized to release oxygen.
3. Microbial Metabolism:
  - Different microbes use various electron acceptors based on redox potential (e.g., nitrate, sulfate).
4. Environmental Redox Conditions:
  - Aerobic soils/waters: High redox potential (oxidizing).
  - Anaerobic soils/waters: Low or negative redox potential (reducing).

#### Applications:

- Wastewater treatment: Monitoring redox potential helps optimize aerobic/anaerobic processes.
- Soil science: Eh indicates soil aeration and nutrient availability.
- Electrochemistry and batteries: Redox potential is crucial in designing galvanic cells.
- Food and fermentation industries: Redox control affects microbial growth and product quality.

#### Factors Affecting Redox Potential:

- pH of the medium
- Temperature
- Concentration of redox species
- Presence of oxidizing/reducing agents

## V. PRINCIPLES OF DIFFUSION AND OSMOSIS – HYPO, HYPER AND ISOTONIC SOLUTIONS

### I. DIFFUSION

#### Definition:

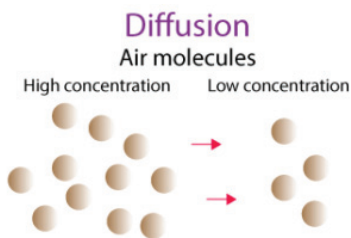
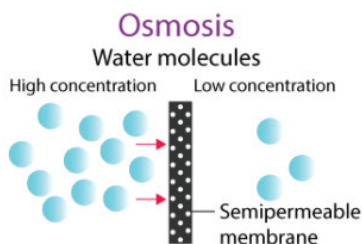
Diffusion is the movement of molecules (solid, liquid, or gas) from a region of higher concentration to a region of lower concentration until equilibrium is reached.

#### Key Features:

- Passive process (no energy required).
- Occurs in gases and liquids.
- Depends on concentration gradient, temperature, and molecular size.

#### Biological Examples:

- Exchange of  $O_2$  and  $CO_2$  in alveoli.
- Movement of glucose and amino acids in tissue fluids.



## II. OSMOSIS

### Definition:

Osmosis is the movement of water molecules across a semipermeable membrane from a region of low solute concentration (high water potential) to a region of high solute concentration (low water potential).

### Key Features:

- Special type of diffusion (only water moves).
- Occurs across cell membranes.
- Essential for water balance in cells.

### Biological Examples:

- Water absorption by root hairs.
- Regulation of water in red blood cells.

### Importance in Biology

- Maintains homeostasis.
- Regulates cell volume.
- Essential for nutrient uptake and waste removal.
- Drives water transport in plants via roots and xylem.

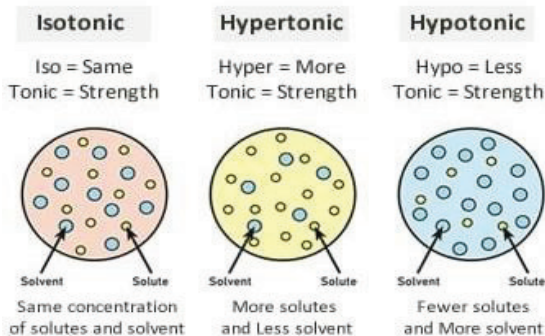
## HYPOTONIC, HYPERTONIC, AND ISOTONIC SOLUTIONS

### Introduction:

Solutions can be classified based on their osmotic pressure or solute concentration relative to the cytoplasm of a cell. The movement of water across a semipermeable membrane is governed by osmosis.

#### 1. Hypotonic Solution:

- Definition: A solution with a lower solute concentration than the cell's cytoplasm.
- Water Movement: Water moves into the cell.
- Effect on Animal Cells: Cell swells and may burst (lysis).
- Effect on Plant Cells: Cell becomes turgid (swollen but protected by the cell wall).
- Example: Distilled water.



## **Types of Solutions**

### **2. Hypertonic Solution:**

- Definition: A solution with a higher solute concentration than the cell's cytoplasm.
- Water Movement: Water moves out of the cell.
- Effect on Animal Cells: Cell shrinks (crenation).
- Effect on Plant Cells: Cell undergoes plasmolysis (cell membrane pulls away from the wall).
- Example: Salt solution (e.g., 10% NaCl).

### **3. Isotonic Solution:**

- Definition: A solution with equal solute concentration as the cell's cytoplasm.
- Water Movement: No net movement of water.
- Effect on Animal Cells: Cell retains normal shape.
- Effect on Plant Cells: Becomes flaccid (not turgid).
- Example: 0.9% NaCl (normal saline).

### **Applications in Biology & Medicine:**

- IV fluids: Must be isotonic to avoid damaging red blood cells.
- Salt preservation: Hypertonic solutions draw water out of bacteria, killing them.
- Agriculture: Over-fertilizing soil creates hypertonic conditions, damaging plant roots.

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## UNIT III: THERMODYNAMICS

### FIRST AND SECOND LAWS OF THERMODYNAMICS

#### 1. FIRST LAW OF THERMODYNAMICS (LAW OF CONSERVATION OF ENERGY)

Statement:

Energy can neither be created nor destroyed, but it can be transformed from one form to another.

Mathematical form:

$$\Delta U = Q - W$$

- $\Delta U$  = Change in internal energy of the system
- $Q$  = Heat supplied to the system
- $W$  = Work done by the system

Explanation:

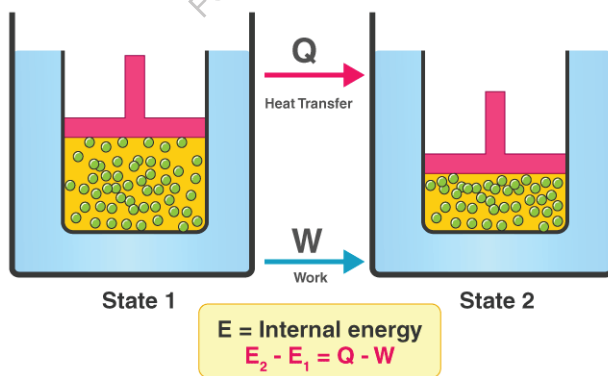
- When a system absorbs heat ( $Q$ ), part of it increases its internal energy ( $\Delta U$ ) and part of it is used to do work ( $W$ ) on the surroundings.
- If  $W$  is positive, the system does work on the surroundings (expansion).
- If  $W$  is negative, work is done on the system (compression).

Special Cases:

- Isothermal process ( $\Delta U = 0$ ):  
 $Q = W$
- Adiabatic process ( $Q = 0$ ):  
 $\Delta U = -W$
- Isobaric process (constant pressure):  
 $Q = \Delta U + P\Delta V$
- Isochoric process ( $\Delta V = 0$ ,  $W = 0$ ):  
 $Q = \Delta U$

Limitations:

- It does not tell about the direction of the process.
- It does not indicate the feasibility or spontaneity of a process.



First Law of Thermodynamics

## 2. SECOND LAW OF THERMODYNAMICS

### 1. Kelvin-Planck Statement (Heat engines):

It is impossible to construct an engine that operates in a cycle and converts all heat absorbed from a heat reservoir into an equivalent amount of work.  
*No engine is 100% efficient.*

### 2. Clausius Statement (Refrigerators):

It is impossible to transfer heat from a colder body to a hotter body without the aid of external work.

### Concept of Entropy (S):

- Entropy is a measure of disorder or randomness.
- For a reversible process:  
 $\Delta S = Q_{\text{rev}}/T$
- For the universe (isolated system), entropy always increases in a spontaneous process:  
 $\Delta S_{\text{total}} \geq 0$

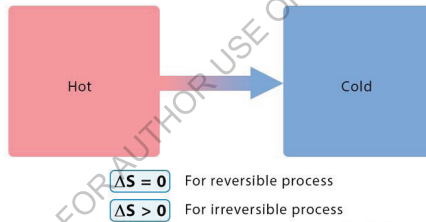
### Implications:

- Explains the direction of natural processes.
- Heat flows spontaneously from hot to cold, not vice versa.
- Perpetual motion machines of the second kind (PMM2) are impossible.

### Second Law of Thermodynamics

The entropy (S) of any natural and spontaneous process either increases or remains constant

Example: Heat flow from a hot body to a cold body



### Applications:

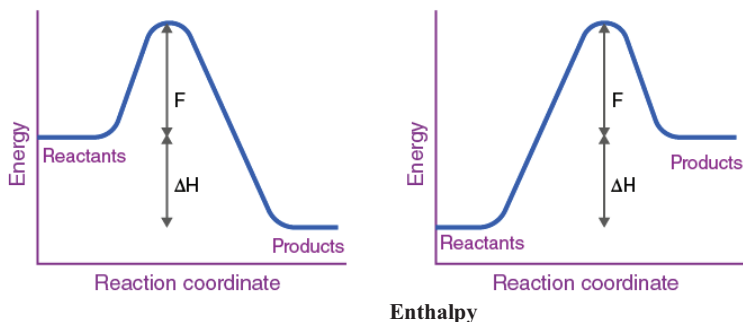
- Heat engines and refrigerators.
- Biological energy transformations (ATP hydrolysis).
- Metabolism and thermoregulation in living organisms.

## BIOLOGICAL APPLICATIONS OF ENTHALPY, FREE ENERGY, ACTIVATION ENERGY, UNAVAILABLE ENERGY AND ENTROPY

### 1. BIOLOGICAL APPLICATIONS OF ENTHALPY

#### 1. Definition of Enthalpy (H)

- Enthalpy is the total heat content of a system.
- It is the sum of internal energy (U) and the product of pressure (P) and volume (V):  
 $H = U + PV$



## 2. Enthalpy Change ( $\Delta H$ )

- $\Delta H$  is the heat absorbed or released in a reaction at constant pressure.
- Exothermic reaction:  $\Delta H < 0$  (releases heat)
- Endothermic reaction:  $\Delta H > 0$  (absorbs heat)

### Applications in Biology

#### 1. Metabolic Reactions

- Cellular respiration is exothermic (releases energy):
  - $\text{Glucose} + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Energy}$  ( $\Delta H < 0$ )
- Photosynthesis is endothermic (requires energy):
  - $\text{CO}_2 + \text{H}_2\text{O} + \text{Light} \rightarrow \text{Glucose} + \text{O}_2$  ( $\Delta H > 0$ )

#### 2. ATP Hydrolysis

- $\text{ATP} \rightarrow \text{ADP} + \text{Pi}$  (Inorganic phosphate)
- Releases a large amount of energy ( $\Delta H < 0$ ), used in:
  - Muscle contraction
  - Active transport
  - Biosynthesis of macromolecules

#### 3. Enzyme-Catalyzed Reactions

- Enthalpy helps in understanding the energy barrier (activation energy).
- Enzymes lower the activation enthalpy to increase reaction rates.

#### 4. Thermoregulation in Animals

- Endothermic animals (like birds and mammals) maintain body temperature by:
  - Exothermic metabolic reactions
  - Regulation of heat loss and production

#### 5. Protein Folding and Denaturation

- Protein folding involves changes in enthalpy and entropy.
- Denaturation by heat alters  $\Delta H$ , breaking hydrogen bonds and other interactions.

#### 6. Membrane Phase Transitions

- Biological membranes undergo phase changes (gel  $\leftrightarrow$  fluid) depending on  $\Delta H$ .
- This affects membrane fluidity and function.

#### 7. Drug-Receptor Interactions

- Binding of a drug to its receptor involves enthalpic changes:
  - Hydrogen bonding
  - Van der Waals forces
  - Ionic interactions

## 2. BIOLOGICAL APPLICATIONS OF FREE ENERGY

### Introduction

- Free energy (Gibbs free energy) is the energy available to do work in a biological system.
- It determines the feasibility and direction of biochemical reactions.
- Represented as  $\Delta G$  (change in free energy).



$$\Delta G = \Delta H - T\Delta S$$

$\Delta G$  : Change in Gibbs free energy

$\Delta H$  : Change in Enthalpy

$\Delta S$  : Change in Entropy

### Key Concepts

- $\Delta G < 0 \rightarrow$  Spontaneous (exergonic) reaction (energy-releasing)
- $\Delta G > 0 \rightarrow$  non-spontaneous (endergonic) reaction (energy-requiring)
- $\Delta G = 0 \rightarrow$  Reaction is at equilibrium

### Biological Significance

#### 1. ATP and Energy Coupling

- Adenosine Triphosphate (ATP) is the universal energy currency.
- Hydrolysis of ATP releases free energy:  
 $\text{ATP} \rightarrow \text{ADP} + \text{Pi}$ ;  $\Delta G \approx -7.3 \text{ kcal/mol}$
- Used to drive endergonic processes like:
  - Muscle contraction
  - Active transport ( $\text{Na}^+/\text{K}^+$  pump)
  - Biosynthesis of macromolecules

#### 2. Enzyme Function

- Enzymes lower the activation energy ( $E_a$ ), not  $\Delta G$ .
- They help in channeling free energy efficiently through metabolic pathways.
- Example: Glycolysis enzymes convert glucose to pyruvate releasing usable energy.

#### 3. Cellular Respiration

- $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy (ATP)}$ ;  $\Delta G \approx -686 \text{ kcal/mol}$
- This large negative  $\Delta G$  shows the process is highly exergonic.
- Energy is captured in ATP and NADH.

#### 4. Photosynthesis

- An endergonic process requiring sunlight ( $\Delta G > 0$ ):
  - $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
- Energy from photons is converted into chemical free energy (glucose).

#### 5. Metabolic Pathway Regulation

- Cells regulate metabolic pathways using  $\Delta G$  values:
  - Reactions with large  $-\Delta G$  are irreversible and regulated steps.
  - Helps maintain homeostasis.

#### 6. Protein Folding and Stability

- Proper protein conformation is achieved when  $\Delta G$  is minimum.
- Misfolding (abnormal  $\Delta G$ ) can lead to diseases like Alzheimer's.

### 7. Membrane Transport

- Active transport requires ATP (positive  $\Delta G$ ) – e.g., ion pumps.
- Passive transport occurs spontaneously (negative  $\Delta G$ ) – e.g., diffusion.

### 8. Signal Transduction

- Activation of receptors often involves ATP hydrolysis or GTP binding.
- Free energy drives conformational changes in signaling proteins.

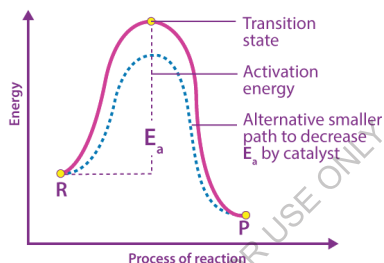
### 9. Biosynthesis of Macromolecules

- Formation of DNA, RNA, proteins is endergonic.
- Coupled with ATP/GTP hydrolysis to make the process energetically favorable.

## 3. BIOLOGICAL APPLICATIONS OF ACTIVATION ENERGY

### Definition of Activation Energy

- Activation energy ( $E_a$ ) is the minimum amount of energy required for a chemical reaction to occur.
- It represents the energy barrier that reactants must overcome to form products.



Activation Energy

### Biological Significance of Activation Energy

#### 1. Role in Metabolism

- Metabolic reactions (e.g., cellular respiration, photosynthesis) require specific activation energies.
- Without catalysts, many reactions in living cells would proceed too slowly to sustain life.

#### 2. Enzyme Action

- Enzymes are biological catalysts that lower the activation energy of biochemical reactions.
- By reducing  $E_a$ , enzymes increase the rate of reactions without being consumed.
- Example: Amylase lowers the  $E_a$  needed to break down starch into sugars.

#### 3. Temperature Regulation in Organisms

- At normal physiological temperatures (37°C in humans), most reactions would not proceed efficiently without reduced activation energy.
- Enzymes ensure that even at mild temperatures, necessary biochemical reactions occur rapidly.

#### 4. Drug Design and Enzyme Inhibition

- Drugs can be designed to bind enzymes and modify their activity by affecting the activation energy.
- Competitive inhibitors increase  $E_a$ , reducing reaction rates – used in antibiotics and cancer treatments.

#### 5. Energy Transfer Reactions

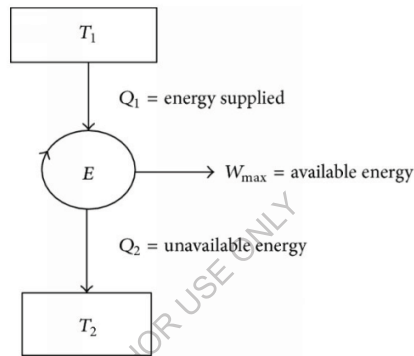
- ATP hydrolysis is a key biological reaction with a moderate activation energy, which allows it to release energy efficiently when catalyzed by enzymes like ATPase.

6. Photosynthesis
  - Light energy absorbed by chlorophyll helps overcome activation energy for splitting water (photolysis) and reducing  $\text{CO}_2$  in the Calvin cycle.
7. Cell Signaling and Biochemical Pathways
  - Signal transduction often involves enzyme cascades.
  - Each step involves overcoming activation energy barriers, usually with the help of kinases and phosphatases.

#### 4. BIOLOGICAL APPLICATIONS OF UNAVAILABLE ENERGY

Definition of Unavailable Energy

- Unavailable energy refers to the part of total energy in a system that cannot be used to do biological work.
- It is often associated with entropy (disorder) in a biological system.



#### Unavailable Energy

Thermodynamic Basis

- According to the Second Law of Thermodynamics, in any energy transformation: Some energy becomes unavailable to do work, often lost as heat.
- Gibbs Free Energy Equation:
 
$$\Delta G = \Delta H - T\Delta S$$
  - $\Delta G$  = change in free energy (available for work)
  - $\Delta H$  = total energy (enthalpy)
  - $T\Delta S$  = energy lost due to entropy (unavailable energy)

Biological Relevance

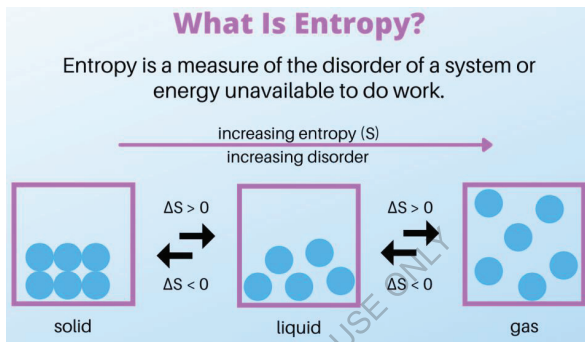
1. Cellular Respiration:
  - Glucose breakdown releases energy.
  - Only ~40% is captured as ATP (usable).
  - The remaining ~60% is lost as heat (unavailable energy).
  - Heat helps in maintaining body temperature in warm-blooded animals.
2. Metabolism:
  - Not 100% efficient.
  - Some energy is always lost as waste heat due to molecular disorder.
3. Homeostasis:
  - Unavailable energy (as heat) contributes to thermoregulation in mammals and birds.
4. Ecosystems:
  - In food chains, only ~10% of energy is passed to the next trophic level.
  - The rest becomes unavailable (lost as heat or waste) due to entropy.
5. Muscle Contraction:

- ATP hydrolysis drives contraction.
  - Energy lost as heat during muscle activity is unavailable but helps in thermogenesis.
6. Enzyme Reactions:
- Energy transformations during enzyme-catalyzed reactions always result in some unavailable energy (entropy).

## 5. BIOLOGICAL APPLICATIONS OF ENTROPY

### Definition of Entropy

- Entropy (S) is a measure of disorder or randomness in a system.
- In thermodynamics, it reflects the number of microscopic configurations corresponding to a macroscopic state.



### Relevance of Entropy in Biology

Biological systems, though highly ordered, follow the second law of thermodynamics: "In any spontaneous process, the total entropy of the universe increases."

#### 1. Protein Folding and Unfolding

- Unfolded proteins = High entropy (many conformations).
- Folded proteins = Low entropy (specific conformation).
- Hydrophobic effect helps fold proteins despite local decrease in entropy.
- Overall entropy (system + surroundings) increases due to water molecules becoming more disordered.

#### 2. Enzyme-Substrate Interactions

- Binding of enzyme and substrate reduces entropy (more order).
- Entropy compensation occurs through release of water molecules or conformational changes.
- Thermodynamic balance of  $\Delta G = \Delta H - T\Delta S$  guides binding affinity.

#### 3. DNA Double Helix Formation

- DNA strands → Single-stranded state = more entropy.
- When DNA strands pair (anneal), entropy decreases locally.
- But overall entropy may increase due to water release and hydrogen bond formation.

#### 4. Cell Metabolism

- Catabolic reactions (e.g., glucose breakdown) increase entropy by breaking down complex molecules into simpler ones.
- Anabolic reactions (e.g., protein synthesis) decrease entropy locally, requiring input of energy (ATP).

#### 5. Osmosis and Diffusion

- Movement of solutes or water from high concentration to low increases entropy.
- Random distribution of molecules = high entropy state.

- Entropy drives spontaneous diffusion across membranes.
6. Homeostasis and Entropy
- Living organisms maintain order (low entropy) through constant energy input (e.g., from food or sunlight).
  - Entropy is exported to surroundings (e.g., as heat) to maintain internal order.
7. Evolution and Entropy
- Evolution may seem like increasing order, but:
    - Local entropy (e.g., cellular complexity) can decrease.
    - Global entropy (total system + environment) still increases.
  - Sun provides energy to drive processes that reduce entropy locally.
8. Entropy in Ecosystems
- Energy transfer between trophic levels leads to loss of usable energy (as heat).
  - This increases entropy in the ecosystem.
  - Explains 10% energy transfer rule.

#### Key Equation

Gibbs Free Energy Equation:

$$\Delta G = \Delta H - T\Delta S$$

- $\Delta G$ : Gibbs free energy
- $\Delta H$ : Enthalpy change
- T: Temperature (Kelvin)
- $\Delta S$ : Entropy change

Spontaneous process:  $\Delta G < 0$ , which often involves  $\Delta S > 0$

## THERMODYNAMICS OF PASSIVE AND ACTIVE TRANSPORT

### Introduction

Cell membranes regulate the movement of molecules via passive and active transport. Thermodynamics helps understand the energy changes involved in these processes.

### 1. Passive Transport

#### Definition:

Movement of substances along the concentration gradient (from high to low concentration) without energy (ATP) input.

#### Types:

- Simple diffusion (e.g.,  $O_2$ ,  $CO_2$ )
- Facilitated diffusion (e.g., glucose via carrier proteins)
- Osmosis (movement of water)

#### Thermodynamic Characteristics:

- $\Delta G$  (Gibbs Free Energy) is negative  $\Rightarrow$  spontaneous
- No external energy required
- Driven by:
  - Concentration gradient
  - Electrochemical gradient (for ions)
- Entropy increases (disorder increases as molecules spread out)

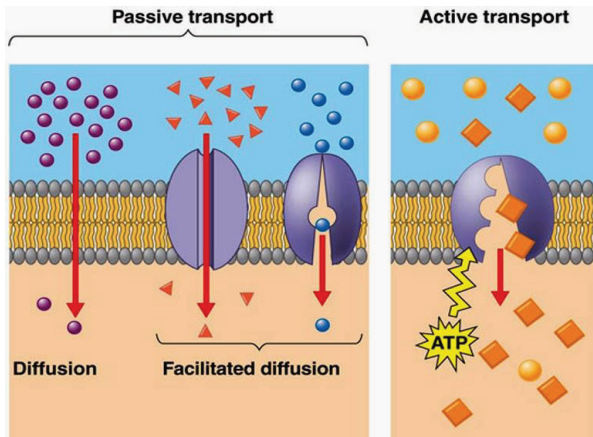
#### Equation:

$$\Delta G = RT \ln([C]_i/[C]_o)$$

#### Where:

- $\Delta G$ : Change in free energy
- R: Gas constant
- Y: Temperature (Kelvin)
- $[C]_i, [C]_o$ : Concentration inside and outside

If  $\Delta G < 0$ : Passive transport is favorable.



### Passive and Active Transport

#### 2. Active Transport

Definition:

Movement of molecules against the concentration gradient (from low to high concentration) using metabolic energy (ATP).

Types:

- Primary Active Transport – Uses ATP directly (e.g.,  $\text{Na}^+/\text{K}^+$  pump)
- Secondary Active Transport – Uses ion gradient set by primary transport (e.g., glucose- $\text{Na}^+$  symporter)

Thermodynamic Characteristics:

- $\Delta G$  is positive  $\Rightarrow$  non-spontaneous
- Requires external energy (ATP hydrolysis or ion gradient)
- Entropy may decrease (more order created)

ATP Contribution:

- ATP hydrolysis:  
 $\text{ATP} \rightarrow \text{ADP} + \text{P}_i + \Delta G \approx -30.5 \text{ kJ/mol}$   
 This negative  $\Delta G$  is coupled with the positive  $\Delta G$  of transporting ions.

◆ Equation:

$$\Delta G = RT \ln([C]_i/[C]_o) + zF\Delta\Psi$$

Where:

- $z$ : ion charge
- $F$ : Faraday's constant
- $\Delta\Psi$ : membrane potential
- Other terms as before.

If total  $\Delta G > 0$ , ATP hydrolysis compensates.

## UNIT IV: SIGNALING AND KINETICS

### **ENZYME ACTION: MICHAELIS-MENTON EQUATION - $V_{MAX}$ - $K_M$ - LINE WEAVER BURK PLOT**

#### 1. Introduction to Enzymes

- Enzymes are biological catalysts made of proteins.
- They speed up biochemical reactions without being consumed.
- Most enzymes are highly specific to substrates.

#### 2. Mechanism of Enzyme Action

##### A. Formation of Enzyme-Substrate Complex (ES complex)

- The enzyme binds to the substrate to form an enzyme-substrate complex.
- This interaction reduces the activation energy required for the reaction.

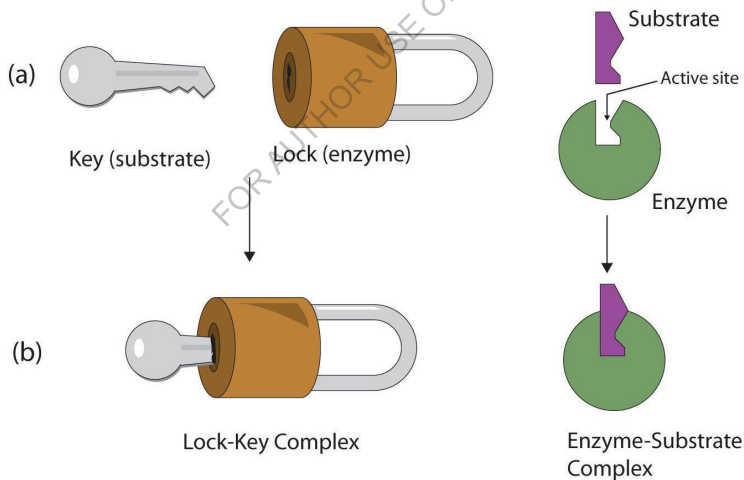
##### B. Models of Enzyme Action

###### 1. Lock and Key Model (Fischer, 1894)

- The substrate fits exactly into the active site of the enzyme.
- Like a key fits into a specific lock.

###### 2. Induced Fit Model (Koshland, 1958)

- The enzyme's active site undergoes a conformational change to fit the substrate better.



### **Lock and Key Model**

#### 3. Stages of Enzyme Action

1. Substrate binds to the active site.
2. Formation of enzyme-substrate complex (ES).
3. Conversion to enzyme-product complex (EP).
4. Release of the product and regeneration of free enzyme.

#### 4. Factors Affecting Enzyme Action

| Factor                  | Effect                                                                                         |
|-------------------------|------------------------------------------------------------------------------------------------|
| Temperature             | Increases activity up to an optimum point (usually ~37°C); high temperatures denature enzymes. |
| pH                      | Each enzyme has an optimal pH; deviation reduces activity.                                     |
| Substrate concentration | Rate increases until enzyme saturation is reached (Michaelis-Menten kinetics).                 |
| Enzyme concentration    | Directly proportional to reaction rate (if substrate is in excess).                            |
| Inhibitors              | Reduce enzyme activity (see below).                                                            |

#### 5. Enzyme Inhibition

- Competitive Inhibition: Inhibitor resembles substrate and binds to active site.
- Non-competitive Inhibition: Inhibitor binds elsewhere, altering enzyme shape.
- Irreversible Inhibition: Inhibitor permanently binds or denatures the enzyme.

#### 6. Enzyme Kinetics – Michaelis-Menten Equation

- $V = (V_{max} \times [S]) / (K_m + [S])$ 
  - $V$  = Reaction rate
  - $V_{max}$  = Maximum rate
  - $[S]$  = Substrate concentration
  - $K_m$  = Michaelis constant (substrate concentration at  $\frac{1}{2} V_{max}$ )

#### 7. Cofactors and Coenzymes

- Cofactors: Non-protein helpers (e.g., metal ions like  $Zn^{2+}$ ,  $Mg^{2+}$ ).
- Coenzymes: Organic molecules (e.g.,  $NAD^+$ , FAD, vitamins).
- Apoenzyme + Cofactor = Holoenzyme (active form).

#### 8. Importance of Enzyme Action

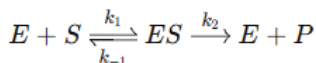
- Essential for metabolism, digestion, DNA replication, energy production, etc.
- Used in industry (e.g., brewing, dairy, detergents), and medicine (e.g., enzyme therapy).

### MICHAELIS-MENTEN EQUATION

#### Introduction

- The Michaelis-Menten equation describes how the rate of an enzyme-catalyzed reaction depends on the substrate concentration.
- Proposed by Leonor Michaelis and Maud Menten in 1913.

#### Basic Enzyme Reaction



#### Where:

- E = Enzyme
- S = Substrate
- ES = Enzyme-substrate complex
- P = Product
- $k_1$ ,  $k_{-1}$ ,  $k_2$  = rate constants

## Michaelis - Menten Equation

$$v = \frac{V_{max}[S]}{K_m + [S]}$$

Where:

- $v$  = Initial reaction velocity
- $V_{max}$  = Maximum reaction velocity (when enzyme is saturated)
- $[S]$  = Substrate concentration
- $K_m$  = Michaelis constant (substrate concentration at half  $V_{max}$ )

Significance of Parameters

- $V_{max}$ : Indicates enzyme's maximum efficiency.
- $K_m$ :
  - Low  $K_m \rightarrow$  High affinity between enzyme and substrate
  - High  $K_m \rightarrow$  Low affinity

## INTERPRETATION OF $V_{max}$ AND $K_m$

### 1. Introduction

In enzyme kinetics, Michaelis-Menten equation describes the rate of enzymatic reactions:

$$v = \frac{V_{max} \cdot [S]}{K_m + [S]}$$

Where:

- $v$  = Initial reaction velocity
- $V_{max}$  = Maximum reaction velocity
- $[S]$  = Substrate concentration
- $K_m$  = Michaelis constant

### 2. $V_{max}$ (Maximum Velocity)

- Definition: The maximum rate of the reaction when all active sites of the enzyme are saturated with substrate.
- Represents:
  - The maximum enzymatic activity.
  - Achieved at very high substrate concentrations.
- Units: Usually expressed in  $\mu\text{mol}/\text{min}$  or moles/litre/second ( $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$ ).
- Interpretation:
  - High  $V_{max} \rightarrow$  fast enzyme activity at saturation.
  - Low  $V_{max} \rightarrow$  slow maximum activity.

### 3. $K_m$ (Michaelis Constant)

- Definition: The substrate concentration at which the reaction velocity is half of  $V_{max}$ .
- Represents:
  - The affinity of the enzyme for its substrate.
- Units: Concentration (e.g., mM or mol/L).
- Interpretation:
  - Low  $K_m \rightarrow$  High affinity (enzyme binds substrate easily).
  - High  $K_m \rightarrow$  Low affinity (requires more substrate to reach  $\frac{1}{2} V_{max}$ ).

## LINEWEAVER–BURK PLOT (DOUBLE RECIPROCAL PLOT)

### Definition:

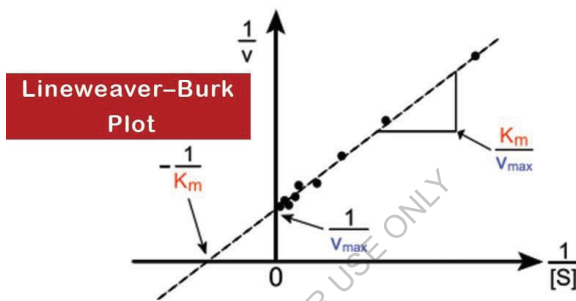
The Lineweaver–Burk plot is a graphical representation of enzyme kinetics, obtained by taking the reciprocal of both sides of the Michaelis–Menten equation. It is also called the double reciprocal plot.

Michaelis–Menten Equation:

$$v = \frac{V_{\max}[S]}{K_m + [S]}$$

Where:

- $v$  = initial reaction velocity
- $V_{\max}$  = maximum velocity
- $[S]$  = substrate concentration
- $K_m$  = Michaelis constant



Double Reciprocal Form:

Taking reciprocals on both sides:

$$\frac{1}{v} = \frac{K_m}{V_{\max}} \cdot \frac{1}{[S]} + \frac{1}{V_{\max}}$$

This is in the form:

$$Y = mx + c$$

Where:

- $y = 1/v$
- $x = 1/[S]$
- Slope ( $m$ ) =  $K_m/V_{\max}$
- y-intercept =  $1/V_{\max}$
- x-intercept =  $-1/K_m$

Plot Characteristics:

- A straight line is obtained when  $1/v$  is plotted against  $1/[S]$ .
- Slope gives  $K_m/V_{\max}$
- Y-intercept =  $1/V_{\max}$
- X-intercept =  $-1/K_m$

Applications:

- Determines  $K_m$  and  $V_{max}$  accurately.
- Used to analyze enzyme inhibition:
  - Competitive inhibition: slope increases, y-intercept same
  - Non-competitive inhibition: y-intercept increases, x-intercept same
  - Uncompetitive inhibition: parallel lines, both intercepts change

Limitations:

- More error-prone at low substrate concentrations.
- Overemphasizes the data points with low  $[S]$ , where measurement errors are more likely.

## ACTION POTENTIAL – REFRACTORY PERIOD – SYNAPTIC POTENTIAL

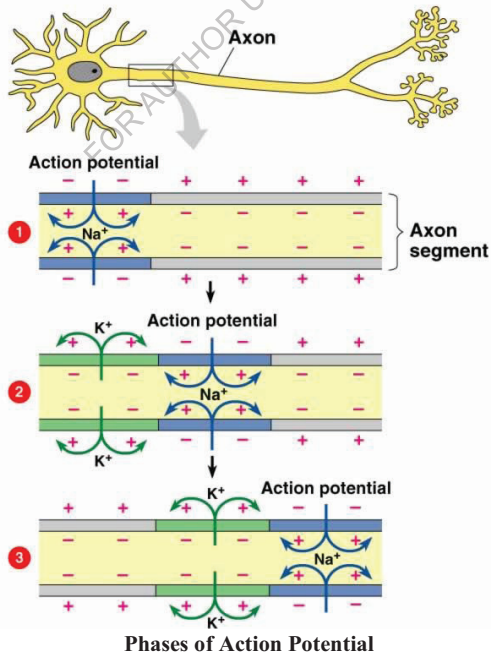
### ACTION POTENTIAL

**Definition:**

An action potential is a rapid, temporary change in the electrical membrane potential of a neuron (or muscle cell) that travels along the axon, enabling nerve signal transmission.

Resting Membrane Potential:

- Resting potential:  $\sim -70$  mV
- Maintained by:
  - $\text{Na}^+/\text{K}^+$  ATPase pump (3  $\text{Na}^+$  out, 2  $\text{K}^+$  in)
  - Selective permeability of the membrane (more permeable to  $\text{K}^+$ )
  - Presence of negatively charged proteins inside the cell



### Phases of Action Potential:

1. Resting State
  - Voltage-gated channels closed.
  - Membrane potential is at  $-70$  mV.
2. Depolarization
  - Stimulus causes  $\text{Na}^+$  channels to open.
  - $\text{Na}^+$  rushes into the cell.
  - Membrane potential becomes positive ( $\sim +30$  mV).
3. Repolarization
  - $\text{Na}^+$  channels close.
  - $\text{K}^+$  channels open.
  - $\text{K}^+$  rushes out of the cell.
  - Membrane returns to negative potential.
4. Hyperpolarization
  - $\text{K}^+$  channels remain open longer.
  - Membrane becomes more negative than resting ( $\sim -80$  mV).

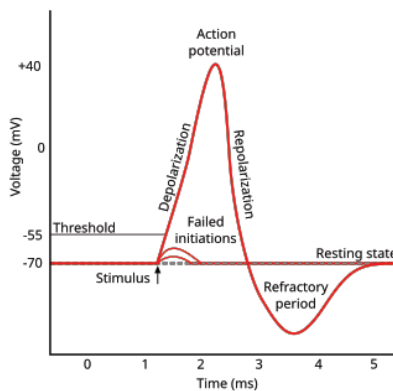
### REFRACTORY PERIOD

#### Definition:

The refractory period is the short period of time after an action potential during which a neuron or muscle fiber is unable or less able to respond to a second stimulus.

#### Types of Refractory Periods:

1. Absolute Refractory Period:
  - Time during which no new action potential can be generated, no matter how strong the stimulus is.
  - Occurs during the depolarization and early repolarization phase of the action potential.
  - Due to inactivation of voltage-gated  $\text{Na}^+$  channels.
2. Relative Refractory Period:
  - Time during which a stronger-than-normal stimulus is needed to initiate a second action potential.
  - Occurs during the late repolarization and hyperpolarization phase.
  - Some  $\text{Na}^+$  channels reset, but  $\text{K}^+$  channels are still open, making it harder to depolarize the membrane.



**The refractory period**

Physiological Basis:

- $\text{Na}^+$  channel inactivation is the main cause of the absolute refractory period.
- Continued  $\text{K}^+$  efflux contributes to the relative refractory period.

Significance:

- Ensures unidirectional propagation of nerve impulses.
- Limits the frequency of action potentials.

## SYNAPTIC POTENTIAL

Definition:

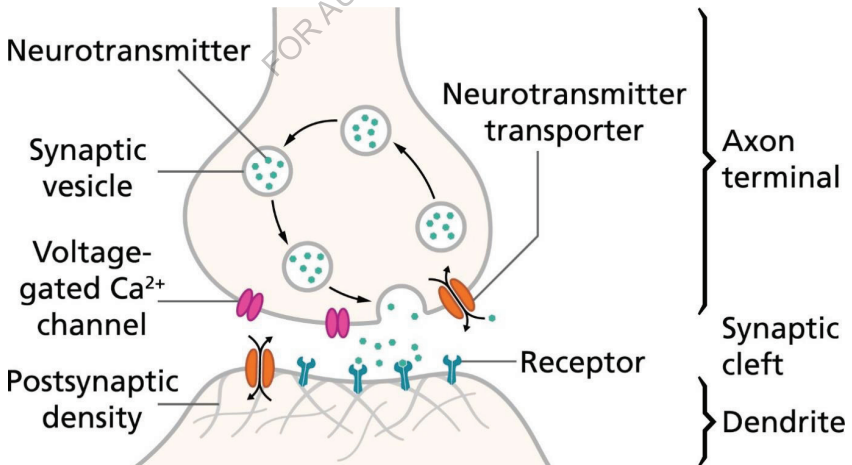
Synaptic potential refers to the change in membrane potential of the postsynaptic neuron resulting from the action of neurotransmitters released at a synapse.

Types of Synaptic Potential:

1. Excitatory Postsynaptic Potential (EPSP):
  - Caused by depolarization of the postsynaptic membrane.
  - Makes the neuron more likely to fire an action potential.
  - Common neurotransmitter: Glutamate, Acetylcholine.
  - Ion movement:  $\text{Na}^+$  enters the postsynaptic neuron.
2. Inhibitory Postsynaptic Potential (IPSP):
  - Caused by hyperpolarization of the postsynaptic membrane.
  - Makes the neuron less likely to fire an action potential.
  - Common neurotransmitter: GABA, Glycine.
  - Ion movement:  $\text{Cl}^-$  enters or  $\text{K}^+$  exits the neuron.

Mechanism of Synaptic Transmission:

1. Arrival of action potential at axon terminal.
2. Opening of voltage-gated  $\text{Ca}^{2+}$  channels.
3.  $\text{Ca}^{2+}$  influx triggers release of neurotransmitters into the synaptic cleft.
4. Neurotransmitters bind to receptors on postsynaptic membrane.
5. Opens ligand-gated ion channels → causes EPSP or IPSP.



**Mechanism of Synaptic Transmission**

Properties:

- Graded (not all-or-none like action potentials).
- Can summate:
  - Spatial Summation: Multiple synapses at different locations stimulate simultaneously.
  - Temporal Summation: Repeated stimuli at the same synapse in quick succession.

Duration:

- Lasts longer than action potentials.
- Duration depends on neurotransmitter and receptor type.

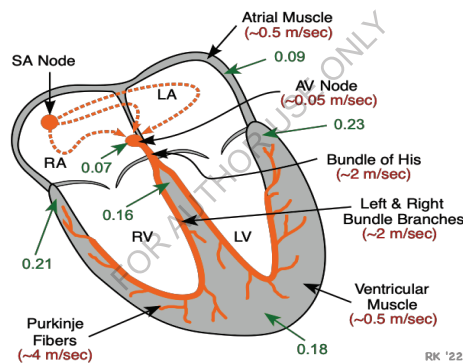
Role:

- Integration of multiple signals to determine whether the postsynaptic neuron will fire an action potential.
- Essential for learning, memory, and neural communication.

## EXCITATION AND CONDUCTION OF HEART BEAT

### Introduction

The heart is a myogenic organ, meaning it can generate its own impulse for contraction. The heartbeat is initiated and conducted by a specialized cardiac conduction system made of modified cardiac muscle fibers (not nerves).



### Excitation and Conduction of Heart Beat

Components of the Cardiac Conduction System

1. Sinoatrial Node (SA Node)
  - Known as the "pacemaker" of the heart
  - Location: Right atrium, near the opening of the superior vena cava
  - Function: Generates impulses (about 70–75 beats/min) that initiate each heartbeat
  - Spreads impulse across both atria → causes atrial contraction
2. Atrioventricular Node (AV Node)
  - Location: Lower part of the right atrium near the interatrial septum
  - Function: Receives impulse from SA node, delays it slightly to allow complete atrial contraction before the ventricles contract
  - Sends impulse to the Bundle of His
3. Bundle of His (Atrioventricular Bundle)
  - Originates from the AV node
  - Runs along the interventricular septum

- Divides into right and left bundle branches that go to respective ventricles
- 4. Purkinje Fibers
  - Terminal branches of the bundle of His
  - Spread through the ventricular myocardium
  - Cause ventricular contraction (systole) starting from the apex upward

#### Sequence of Cardiac Excitation

| Step | Event                                 | Result                              |
|------|---------------------------------------|-------------------------------------|
| 1    | SA node fires                         | Atria contract                      |
| 2    | Impulse reaches AV node               | Brief delay                         |
| 3    | Impulse travels through Bundle of His | Reaches ventricles                  |
| 4    | Impulse spreads via Purkinje fibers   | Ventricles contract (synchronously) |

#### Physiological Features

- Automaticity: Ability to generate impulse without external stimulation (SA node has highest automaticity).
- Rhythmicity: Regular generation of impulses.
- Conductivity: Transmission of impulse through the conduction system.
- Delay at AV Node: Ensures atria empty completely before ventricles contract.

#### Electrocardiogram (ECG) Representation

- P wave – Atrial depolarization
- QRS complex – Ventricular depolarization
- T wave – Ventricular repolarization

## RADIO-LABELING AND TRACER TECHNIQUES

### RADIO-LABELING

#### 1. Definition

Radio-labeling (also called radioactive labeling) is a technique in which radioactive isotopes are attached to molecules (like DNA, RNA, proteins, or drugs) to track or detect them in biological systems.

#### 2. Commonly Used Radioisotopes

| Isotope                         | Radiation Type | Half-life   | Used for labeling               |
|---------------------------------|----------------|-------------|---------------------------------|
| <sup>32</sup> P (Phosphorus-32) | β-emitter      | ~14 days    | DNA, RNA (phosphates)           |
| <sup>3</sup> H (Tritium)        | β-emitter      | ~12 years   | Hormones, nucleotides, drugs    |
| <sup>14</sup> C (Carbon-14)     | β-emitter      | ~5730 years | Organic compounds               |
| <sup>125</sup> I (Iodine-125)   | γ-emitter      | ~60 days    | Proteins, antibodies            |
| <sup>35</sup> S (Sulfur-35)     | β-emitter      | ~87 days    | Proteins (methionine, cysteine) |

#### 3. Purpose of Radio-labeling

- To track the movement of molecules in cells or organisms
- To identify binding sites (e.g., receptor-ligand interaction)
- To study metabolic pathways
- To quantify molecules (e.g., radioimmunoassay)
- To conduct DNA sequencing and hybridization studies

#### 4. Types of Radio-labeling

##### a) Incorporation Labeling

- Radioisotopes are incorporated into the molecule during synthesis (e.g.,  $^{32}\text{P}$  into newly synthesized DNA).

##### b) Post-synthetic Labeling

- Radioisotopes are attached after the molecule is formed (e.g., iodination of proteins).

#### 5. Techniques Using Radio-labeling

- Autoradiography: Visualizes radio-labeled molecules using X-ray film
- Scintillation counting: Quantifies radioactivity
- Radioimmunoassay (RIA): Measures hormones, drugs, and proteins
- Southern/Northern blotting: Detects DNA/RNA with labeled probes
- Pulse-chase experiments: Tracks biosynthetic pathways

#### 6. Advantages

- High sensitivity
- Can detect very small amounts
- Enables real-time tracking

#### 7. Safety Measures

- Use shielding (plexiglass for  $\beta$ -emitters, lead for  $\gamma$ -emitters)
- Wear gloves and lab coats
- Work in designated radioisotope areas
- Use dosimeters to monitor exposure
- Proper disposal of radioactive waste

#### 8. Applications

- Molecular biology research
- Cancer diagnosis and therapy
- Drug development and tracking
- Agricultural studies (e.g., nutrient uptake)

#### 9. Example

##### Labeling DNA with $^{32}\text{P}$ :

- A DNA fragment can be labeled at the 5' end using polynucleotide kinase and  $^{32}\text{P}$ -ATP.
- This labeled DNA can then be used in hybridization experiments.

#### 10. Limitations

- Requires specialized facilities and licensing
- Health hazards due to radiation exposure
- Decay of isotopes limits shelf-life

## TRACER TECHNIQUES

### Definition:

Tracer techniques involve the use of radioactive or stable isotopes (tracers) to track the movement or transformation of elements or molecules within a biological, chemical, or physical system.

### Types of Tracers:

#### 1. Radioactive Isotope Tracers:

- Emit radiation (e.g., alpha, beta, gamma rays).
- Examples:
  - Carbon-14 ( $^{14}\text{C}$ )
  - Phosphorus-32 ( $^{32}\text{P}$ )

- Iodine-131 ( $^{131}\text{I}$ )
  - Tritium ( $^3\text{H}$ )
2. Stable Isotope Tracers:
- Non-radioactive; detected using mass spectrometry.
  - Examples:
    - Carbon-13 ( $^{13}\text{C}$ )
    - Nitrogen-15 ( $^{15}\text{N}$ )
    - Deuterium ( $^2\text{H}$ )

### Principle:

- A tracer (isotope-labeled compound) behaves chemically and biologically like the regular compound.
- It is introduced into a system and detected or measured to study its pathway, location, or transformation.

Instruments Used:

### 1. GEIGER-MÜLLER COUNTER – Detects beta and gamma radiation.

#### Definition

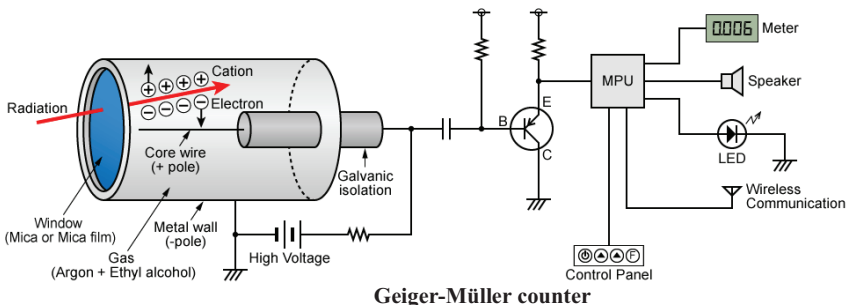
The Geiger-Müller counter is a radiation detection instrument used to detect and measure ionizing radiation, such as alpha particles, beta particles, and gamma rays.

#### Components of a GM Counter

1. Geiger-Müller Tube:
  - A gas-filled tube with a central wire anode and cylindrical cathode.
  - Filled with inert gas (like argon) at low pressure.
2. High Voltage Power Supply:
  - Maintains potential difference (typically 400–900 volts).
3. Counting System:
  - Each detected ionization event produces a click sound or count.
  - Counts are displayed on a digital or analog meter.
4. Amplifier and Speaker:
  - Converts pulses into audible clicks.

#### Working Principle

- When ionizing radiation enters the tube, it ionizes the gas.
- Electrons are attracted to the positive anode, while positive ions move toward the cathode.
- This causes a cascade of ionizations → large current pulse → detected as a "count".



### Applications

- Radiotracer studies in biology and medicine.
- Radiation monitoring in nuclear labs.
- Detection of radioactive contamination.
- Measuring the half-life of radioactive isotopes.
- Checking radiation leaks in industrial settings.

### Advantages

- Simple and portable.
- Inexpensive.
- Can detect low levels of radiation.
- Audible clicks help with quick identification of sources.

### Limitations

- Cannot differentiate between types of radiation.
- Cannot measure radiation energy.
- Saturates at high radiation levels.

## 2. SCINTILLATION COUNTER – Measures radioactive decay using light flashes.

### Definition:

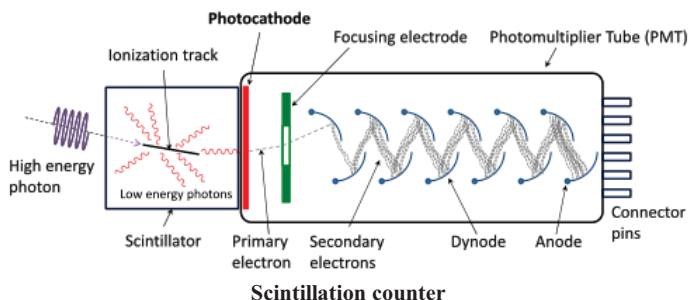
A scintillation counter is a device used to detect and measure ionizing radiation by detecting the flashes of light (scintillations) produced in a scintillating material.

### Principle:

When a radioactive particle hits a scintillating material, it produces a tiny flash of light. This light is converted to an electrical signal by a photomultiplier tube (PMT) and is then counted and measured.

### Main Components:

1. Scintillator: Material that emits light when hit by radiation
  - Solid (e.g., Sodium iodide NaI(Tl))
  - Liquid (used in liquid scintillation counters)
2. Photomultiplier tube (PMT): Converts light to electric signal
3. Amplifier: Strengthens the signal
4. Counter/Display: Displays counts per minute (CPM) or disintegrations per minute (DPM)



### Types of Scintillation Counters:

1. Solid Scintillation Counter: Best for gamma rays
2. Liquid Scintillation Counter: Best for beta emitters like C-14 or H-3

**Advantages:**

- High sensitivity and accuracy
- Can detect low levels of radioactivity
- Suitable for both alpha, beta, and gamma detection

**Applications:**

- Radioimmunoassay
  - Nuclear medicine
  - Environmental radiation monitoring
  - Tracer studies in biology and chemistry
  - Detecting contamination in labs
3. Autoradiography – Visualizes radioactive tracers in tissues/cells.
  4. Mass Spectrometer – Detects and quantifies stable isotopes.

**Applications in Biology:**

1. Metabolic Studies:
  - $^{14}\text{C}$ -glucose used to study glycolysis or cellular respiration.
  - $^{32}\text{P}$  used in studying ATP turnover or nucleic acid synthesis.
2. DNA and Protein Synthesis:
  - $^3\text{H}$ -thymidine used to trace DNA replication.
  - $^3\text{H}$ -leucine or  $^{35}\text{S}$ -methionine to study protein synthesis.
3. Plant Physiology:
  - $^{14}\text{C}$ -labelled  $\text{CO}_2$  in photosynthesis tracking.
  - $^{32}\text{P}$  to study phosphate uptake in roots.
4. Medical Applications:
  - $^{131}\text{I}$  for thyroid function testing.
  - PET scans using radioactive glucose ( $^{18}\text{F}$ -FDG) for cancer detection.
5. Ecological and Environmental Studies:
  - $^{15}\text{N}$  used to track nitrogen cycle in ecosystems.
  - $^2\text{H}$  or  $^{18}\text{O}$  in water movement and climatic studies.

**Advantages:**

- High sensitivity.
- Can trace even minute quantities.
- Non-destructive to the sample (in stable isotope methods).
- Useful in both in vivo and in vitro studies.

**Limitations:**

- Radioactive waste handling and disposal.
- Health hazards due to radiation.
- Requires specialized equipment and trained personnel.
- Short half-life of some isotopes limits long-term studies.

**Safety Measures:**

- Use of lead shielding and protective clothing.
- Regular radiation exposure monitoring (dosimeters).
- Proper storage and disposal of radioactive materials.
- Following institutional and legal safety protocols (e.g., Atomic Energy Regulatory Board guidelines in India).

## UNIT V: INSTRUMENTATION PRINCIPLES

### PRINCIPLES AND APPLICATIONS OF pH METER

#### Introduction

- pH (potential of hydrogen) is a measure of the hydrogen ion concentration  $[H^+]$  in a solution.
- It indicates the acidity or alkalinity of a solution on a scale from 0 to 14.
- pH meter is an electrochemical instrument used for accurate measurement of pH.

#### Principle of pH Meter

##### 1. Electrochemical Basis

- pH is measured based on the Nernst equation, which relates the electrode potential to ion concentration:

$$E = E^0 - \frac{2.303RT}{nF} \log[H^+]$$

Where:

- E = electrode potential,
- $E^0$  = standard electrode potential,
- R = gas constant ( $8.314 \text{ J/mol} \cdot \text{K}$ ),
- T = temperature (in Kelvin),
- n = number of electrons involved ( $n=1$  for  $H^+$ ),
- F = Faraday's constant ( $96,485 \text{ C/mol}$ ),
- $[H^+]$  = hydrogen ion concentration.

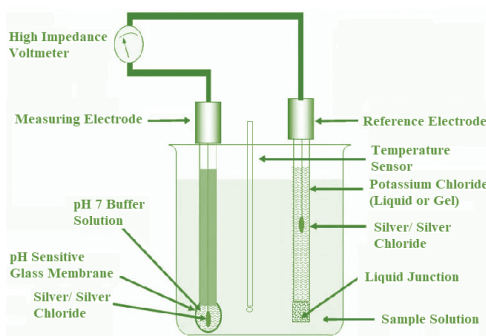
##### 2. Electrodes Used

- Glass Electrode: Sensitive to hydrogen ions; contains a thin membrane responsive to  $H^+$ .
- Reference Electrode: Commonly calomel or silver/silver chloride ( $Ag/AgCl$ ); maintains constant potential.

##### 3. Measurement System

- pH meter measures voltage difference between the glass and reference electrodes.
- This voltage is converted into a pH value using an internal algorithm or calibration curve.

Components of a pH Meter



Components of a pH Meter

1. pH Electrode System
  - Measuring Electrode: Glass membrane responds to  $H^+$ .
  - Reference Electrode: Maintains a stable potential.
2. pH Meter (Electrometer)
  - Converts millivolt (mV) signal to pH.
  - Modern meters are digital with temperature compensation.
3. Temperature Probe
  - Because pH is temperature-dependent, ATC (Automatic Temperature Compensation) ensures accuracy.
4. Calibration Solutions
  - Standard buffer solutions (e.g., pH 4.00, 7.00, 10.00) are used for two- or three-point calibration.

#### Calibration of pH Meter

- Essential for accuracy.
- Two-point calibration: pH 7 and either pH 4 or 10.
- Three-point calibration: pH 4, 7, and 10 for better precision across range.

#### Factors Affecting pH Measurement

- Temperature: Influences electrode response and buffer dissociation.
- Electrode Condition: Fouling or clogging affects performance.
- Ionic Strength of solution.
- Junction Potential: Difference in potential across salt bridge.
- Interference from Ions (e.g., sodium ion error at high pH).

#### Applications of pH Meter

1. Laboratory and Research
  - Monitoring enzyme activity, biochemical reactions.
  - Cell culture media and buffer preparations.
  - Titrations and chemical kinetics.
2. Environmental Science
  - Water quality analysis (e.g., lakes, rivers, groundwater).
  - Soil pH measurement for agriculture.
3. Clinical and Biomedical Fields
  - Measuring pH of biological fluids (e.g., blood, urine).
  - Monitoring pH in cell physiology experiments.
4. Industrial Applications
  - Food Industry: Fermentation control, dairy and beverage quality.
  - Pharmaceuticals: Drug formulation, stability testing.
  - Textile and Leather: Dye bath control.
  - Electroplating: Solution bath pH monitoring.

#### Advantages

- High accuracy and sensitivity.
- Fast and real-time monitoring.
- Portable versions for field use.

### Limitations

- Requires frequent calibration.
- Glass electrode is fragile.
- Measurement errors at extreme pH values.

### Recent Advances

- Solid-state electrodes and ISFET-based (Ion-sensitive field-effect transistor) pH meters.
- Microelectrodes for small-volume or in vivo measurements.
- Wireless and smartphone-integrated digital pH meters.

### pH Meter Examples



pH 1120X (Portable pH Meter)  
(Mettler Toledo)



LAQUATwin pH Meter PH-11 (2 point)



Gastroesophageal impedance pH  
meter (BLU RUNNER)



Portable 913 pH Meter  
(Metrohm)

## PRINCIPLES AND APPLICATIONS OF CENTRIFUGATION

### Introduction

Centrifugation is a separation technique that uses centrifugal force to accelerate the sedimentation of particles in a solution. It is widely used in biological, biochemical, and industrial laboratories for separating particles based on size, shape, density, viscosity of the medium, and rotor speed.

### Principles of Centrifugation

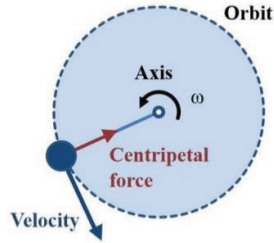
#### 1. Centrifugal Force (RCF)

- The force acting outward on a particle moving in a circle.
- It is proportional to the square of the angular velocity ( $\omega$ ) and the distance ( $r$ ) from the axis of rotation.

$$\text{RCF (g)} = 1.118 \times 10^{-5} \times r \times (\text{RPM})^2$$

Where:

- RCF = Relative Centrifugal Force ( $\times g$ )
- $r$  = radius in cm
- RPM = revolutions per minute



## Principles of Centrifugation

### 2. Sedimentation Rate

- Particles settle at a rate proportional to the centrifugal force and inversely proportional to the frictional resistance.

$$v = \frac{m(1 - \rho/\rho_p) \cdot \omega^2 \cdot r}{f}$$

Where:

- $v$  = sedimentation velocity
- $m$  = particle mass
- $\rho$  = medium density
- $\rho_p$  = particle density
- $f$  = frictional coefficient
- $\omega$  = angular velocity

### 3. Sedimentation Coefficient (Svedberg unit, S)

- Describes how fast a particle sediment.

$$S = \frac{v}{\omega^2 r}$$

- Expressed in Svedberg units (1 S =  $10^{-13}$  seconds).

### Types of Centrifugations

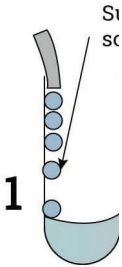
| Type                | Description                                 | Application                             |
|---------------------|---------------------------------------------|-----------------------------------------|
| Differential        | Based on sedimentation rate                 | Cell fractionation                      |
| Density Gradient    | Uses a gradient (e.g., sucrose, CsCl)       | DNA/RNA/protein separation              |
| Rate-Zonal          | Samples layered on top of gradient          | Separation by size and shape            |
| Isopycnic           | Particles move to position of equal density | Separation based on buoyant density     |
| Ultracentrifugation | High-speed rotation (up to 100,000 rpm)     | Macromolecule studies, virus separation |

### Types of Rotors

| Rotor Type      | Features                             | Uses                         |
|-----------------|--------------------------------------|------------------------------|
| Fixed-angle     | Tubes at a fixed angle (e.g., 45°)   | Pellet collection, fast runs |
| Swinging-bucket | Buckets swing to horizontal position | Gradient centrifugation      |
| Vertical        | Tubes remain vertical                | Isopycnic centrifugation     |
|                 |                                      |                              |

## Equipment Components

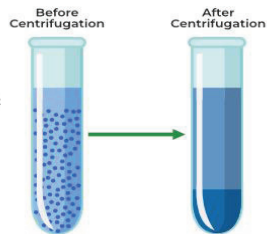
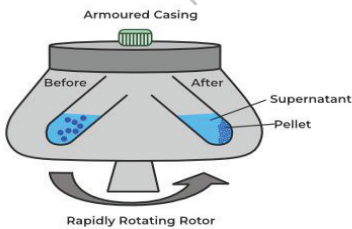
- Rotor
- Chamber with temperature control
- Vacuum system (in ultracentrifuges)
- Tachometer
- Safety interlocks



Small-size  
DNA molecule

Medium-size  
DNA molecule

Large-size  
DNA molecule



## Equipment Components

### Applications of Centrifugation

#### A. Biological Sciences

- Separation of subcellular organelles (nucleus, mitochondria, lysosomes)
- Isolation of nucleic acids and proteins
- Purification of viruses and plasmids
- Differential and density gradient centrifugation in molecular biology



## B. Clinical and Medical Use

- Hematocrit determination
- Separation of blood components (plasma, serum)
- Isolation of diagnostic biomarkers

## C. Industrial Applications

- Milk fat separation
- Wastewater treatment
- Production of vaccines and antibiotics

## D. Research and Development

- Sedimentation analysis of macromolecules
- Study of molecular mass and conformational changes
- Protein-protein interaction studies

## Advanced Techniques

- Analytical Ultracentrifugation (AUC): Determines molecular mass, shape, and interaction.
- Preparative Ultracentrifugation: Prepares large quantities of biological samples.
- Zonal Centrifugation: Used for large sample volumes in gradient.

## Advantages and Limitations

### Advantages:

- High resolution
- Time-efficient
- Versatile (can process a wide range of samples)

### Limitations:

- Requires expensive equipment
- Sample heating at high speeds
- Risk of sample loss due to pellet disturbance

## Safety Precautions

- Proper balancing of tubes
- Regular maintenance
- Use of sealed rotors for hazardous materials
- Adherence to manufacturer's guidelines

# PRINCIPLES AND APPLICATIONS OF CHROMATOGRAPHY

## Introduction

Chromatography is a separation technique used to isolate components in a mixture based on differences in their distribution between two phases:

- Stationary phase: Fixed in place (solid or liquid supported on a solid)
- Mobile phase: Moves through or over the stationary phase (liquid or gas)

## Principles of Chromatography

### 1. Partition Coefficient (K)

$$K = C_s / C_m$$

Where  $C_s$  and  $C_m$  are concentrations in stationary and mobile phases respectively. The separation depends on different K values for each component.

## 2. Differential Migration

Components move at different rates due to differences in:

- Adsorption
- Solubility
- Volatility
- Size or charge

## 3. Rate of Migration

$R_f$  = Distance travelled by solute/Distance travelled by solvent front

## 4. Selectivity and Resolution

- Selectivity ( $\alpha$ ): Ratio of retention factors ( $k'$ ) of two solutes.
- Resolution ( $R_s$ ): Measures how well two solutes are separated.

## 5. Retention Time ( $t_R$ )

Time a compound takes to elute from the column.

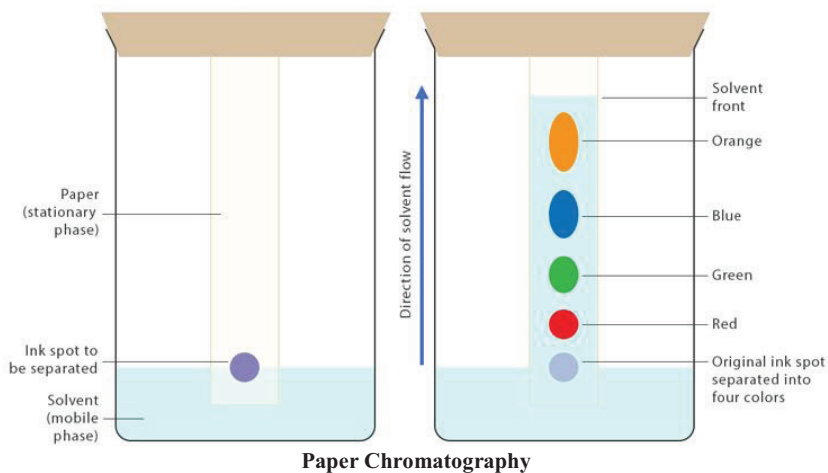
## Types of Chromatography

| Type                                          | Based on                | Example                          |
|-----------------------------------------------|-------------------------|----------------------------------|
| Partition Chromatography                      | Solubility in phases    | Paper chromatography             |
| Adsorption Chromatography                     | Surface interactions    | Thin Layer Chromatography        |
| Ion-Exchange Chromatography                   | Charge interactions     | Protein purification             |
| Size Exclusion Chromatography (SEC)           | Molecular size          | DNA separation                   |
| Affinity Chromatography                       | Specific ligand binding | Antibody purification            |
| Gas Chromatography (GC)                       | Volatility              | Environmental pollutant analysis |
| High Performance Liquid Chromatography (HPLC) | Polarity/interaction    | Drug analysis                    |

## Major Chromatographic Techniques

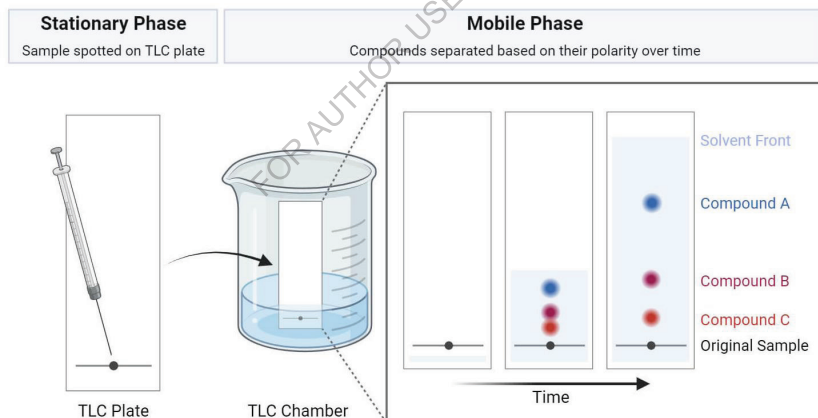
### 1. Paper Chromatography

- Stationary phase: Cellulose paper
- Mobile phase: Solvent
- Principle: Partition
- Used for amino acids, pigments.



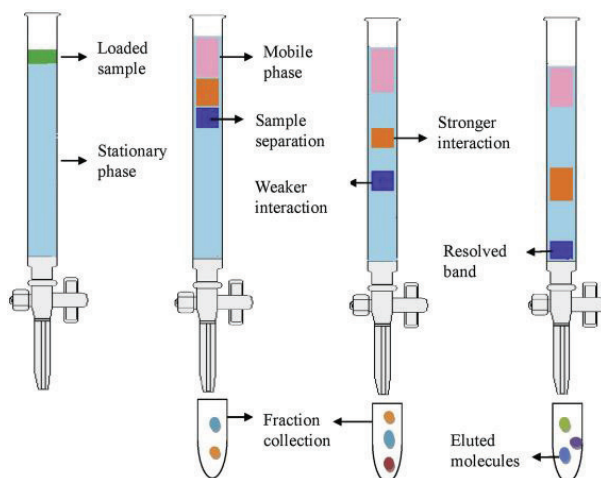
## 2. Thin Layer Chromatography (TLC)

- Stationary: Silica gel/Alumina on glass/plastic
- Quick, qualitative
- Used for purity checks, monitoring reactions.



## 3. Column Chromatography

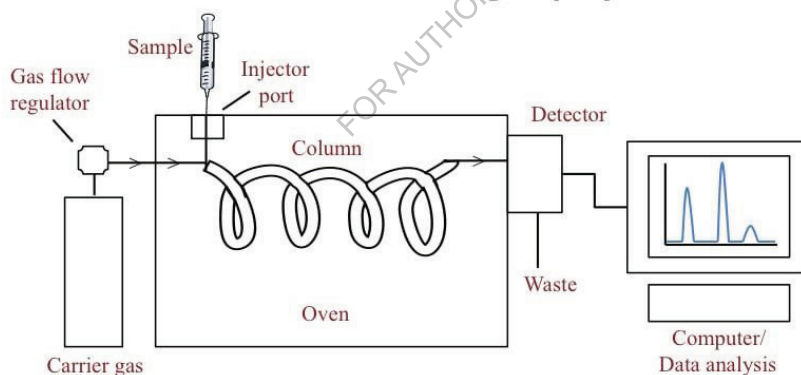
- Column packed with stationary phase
- Gravity or pressure-driven flow
- Used for compound isolation



**Column Chromatography**

#### 4. Gas Chromatography (GC)

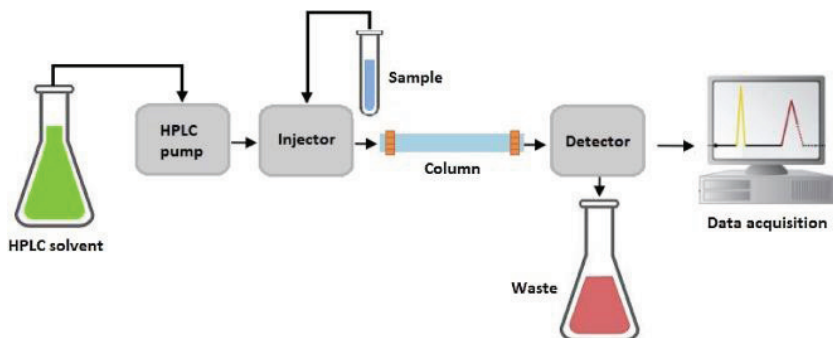
- Mobile phase: Inert gas ( $\text{He}/\text{N}_2$ )
- Stationary phase: Coated column
- High-resolution analysis of volatile compounds



**Gas Chromatography**

#### 5. High Performance Liquid Chromatography (HPLC)

- High-pressure pumps push solvent
- Used for drugs, bioactive compounds
- Reversed-phase HPLC: Non-polar stationary phase



**High Performance Liquid Chromatography**

#### 6. Ion Exchange Chromatography

- Separates based on charge
- Cation or anion exchange resins
- Used for amino acids, proteins, nucleotides

#### 7. Affinity Chromatography

- Specific interaction: Enzyme-substrate, antigen-antibody
- High specificity
- Used in biotechnology

#### 8. Gel Filtration (Size Exclusion)

- Separates based on molecular size
- Larger molecules elute first
- Common for protein desalting, molecular weight estimation

### Applications of Chromatography

#### 1. Pharmaceutical and Biomedical

- Drug purity and content analysis (HPLC)
- Biofluid screening (GC, LC-MS)
- Protein purification (Affinity, SEC)

#### 2. Environmental Analysis

- Detection of pollutants (GC-MS)
- Soil and water contaminants

#### 3. Food and Beverage Industry

- Additive and preservative analysis
- Pesticide residues
- Flavor and fragrance components

#### 4. Forensic Science

- Drug identification
- Explosives and poison analysis
- Ink and dye differentiation (TLC, GC)

#### 5. Molecular Biology

- Nucleic acid and protein separation
- Genetic fingerprinting
- Metabolomics and proteomics

## Advanced Techniques

### 1. LC-MS and GC-MS

- Coupling of chromatography with mass spectrometry
- High sensitivity and identification capabilities

### 2. 2D Chromatography

- Sequential use of two different mechanisms
- Better resolution of complex mixtures

### 3. Chiral Chromatography

- Separation of enantiomers
- Important in drug development

## Limitations

- Sample loss in preparative methods
- Requirement of pure solvents
- Need for expensive equipment (HPLC, GC-MS)
- Not suitable for non-detectable compounds without derivatization

## PRINCIPLES AND APPLICATIONS OF ELECTROPHORESIS

### Introduction

Electrophoresis is a technique used to separate charged biomolecules such as nucleic acids and proteins based on their size, charge, and conformation under the influence of an electric field. It plays a pivotal role in molecular biology, biochemistry, genetics, and forensic science.

### Principle of Electrophoresis

#### 1. Basic Concept:

- When an electric field is applied to a medium containing charged particles, the particles migrate towards the oppositely charged electrode.
- Cations move toward the cathode (–), and anions move toward the anode (+).

#### 2. Factors Influencing Migration:

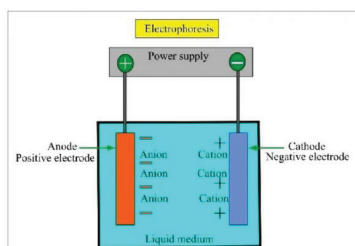
- Net charge of the molecule
- Size and shape of the molecule
- Strength of the electric field
- Type and concentration of buffer
- Nature of the supporting medium

#### 3. Mobility ( $\mu$ ):

$$\mu = v/E$$

where:

- $v$  = velocity of the particle
- $E$  = electric field strength

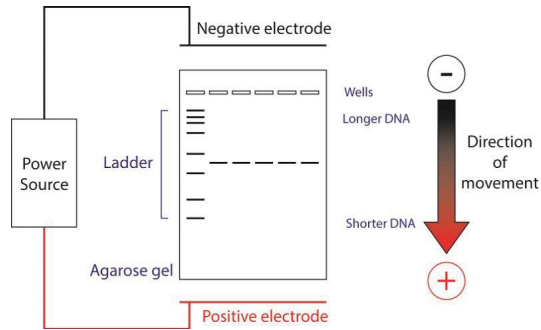


**Principle of Electrophoresis**

## Types of Electrophoresis

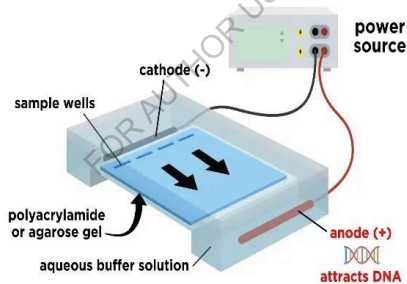
### 1. Gel Electrophoresis

- Agarose Gel Electrophoresis: Used for separation of DNA and RNA.



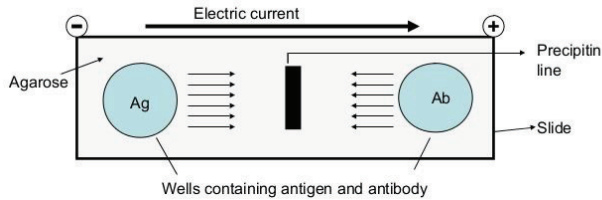
### Gel Electrophoresis

- Polyacrylamide Gel Electrophoresis (PAGE):
  - Native PAGE: Separation based on size and charge.
  - SDS-PAGE: Denatures proteins and separates based on molecular weight.



### Polyacrylamide Gel Electrophoresis

- Capillary Electrophoresis (CE)
  - High-resolution, automated system using thin capillaries.
  - Used for DNA sequencing, forensic analysis, and protein characterization.
- Isoelectric Focusing (IEF)
  - Separates proteins based on their isoelectric points (pI).
  - Proteins stop migrating when the pH of the medium equals their pI.
- Two-Dimensional Electrophoresis (2-DE)
  - Combines IEF and SDS-PAGE.
  - Used for protein profiling in proteomics.
- Immunoelectrophoresis
  - Combines electrophoresis and antigen-antibody reactions for analyzing serum proteins.



### Immunoelectrophoresis

#### Buffers in Electrophoresis

- Buffers maintain a constant pH and provide ions for conduction.
- Common buffers: Tris-acetate, Tris-borate, Tris-glycine, Tris-HCl.

#### Detection and Visualization Techniques

- Ethidium bromide (EtBr) for DNA (fluoresces under UV).
- Coomassie Brilliant Blue and Silver Staining for proteins.
- Fluorescent dyes for sensitive detection.
- Autoradiography for radiolabeled molecules.

#### Applications of Electrophoresis

##### A. Molecular Biology

- DNA fingerprinting and genotyping
- RNA quality assessment
- Plasmid analysis
- PCR product analysis

##### B. Proteomics

- Separation and identification of proteins
- Post-translational modification analysis
- Protein purity checking

##### C. Medical Diagnostics

- Detection of hemoglobin variants (e.g., HbS in sickle cell anemia)
- Serum protein electrophoresis for diagnosing multiple myeloma
- Isoenzyme analysis (e.g., creatine kinase isozymes)

##### D. Forensic Science

- DNA profiling from biological samples
- Paternity testing

##### E. Food and Agriculture

- Protein analysis in food
- Genetic marker analysis in crops

#### Advantages

- High sensitivity and resolution
- Versatility across biomolecule types
- Cost-effective and relatively simple

#### Limitations

- Time-consuming for large-scale analysis
- May require staining or labeling for visualization
- Sample denaturation in some types (e.g., SDS-PAGE)

## PRINCIPLES AND APPLICATIONS OF COLORIMETER AND SPECTROPHOTOMETER

### Introduction

- Colorimeter and spectrophotometer are analytical instruments used to measure the concentration of solutes based on their absorbance of light.
- Both operate on Beer-Lambert's Law, which correlates absorbance with concentration.

### COLORIMETER

#### A. Principle

- Based on Beer's Law and Lambert's Law:

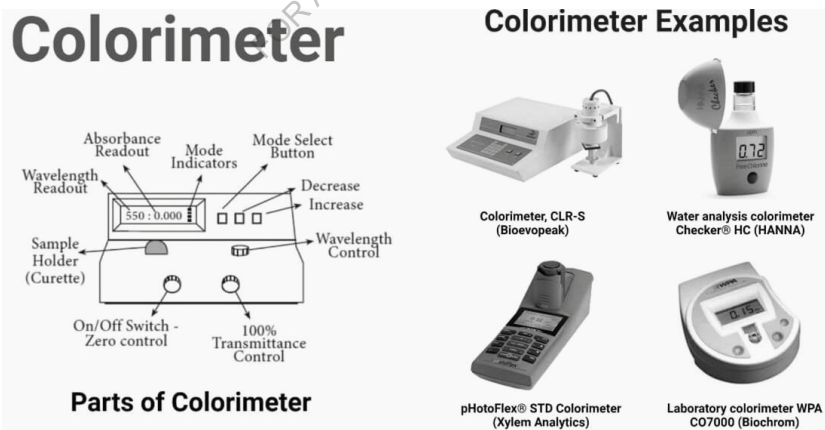
$$A = \log_{10} (I_0/I) = \epsilon \cdot c \cdot l$$

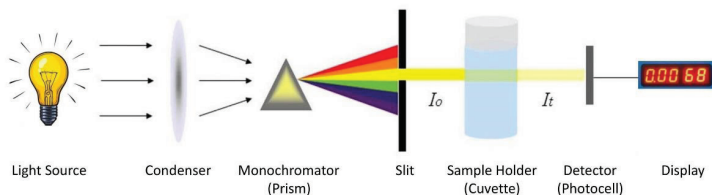
where:

- A = Absorbance
  - $I_0$  = Incident light intensity
  - I = Transmitted light intensity
  - $\epsilon$  = Molar absorptivity
  - c = Concentration
  - l = Path length
- A color filter selects the complementary color of the solution to enhance accuracy.

#### B. Components

1. Light source – Usually a tungsten filament lamp.
2. Filter – Selects a specific wavelength range.
3. Cuvette – Holds the sample (typically 1 cm path length).
4. Photodetector – Measures transmitted light.
5. Meter/Readout – Displays absorbance or % transmittance.





### Working Principle of Colorimeter

#### C. Applications

- Estimation of blood glucose, hemoglobin, urea, protein.
- Clinical biochemistry and food industry testing.
- Used in colorimetric enzyme assays.
- Determination of reaction rates in enzyme kinetics.

#### D. Limitations

- Limited to visible range (400–700 nm).
- Less sensitive and less accurate compared to spectrophotometers.

## SPECTROPHOTOMETER

#### Principle

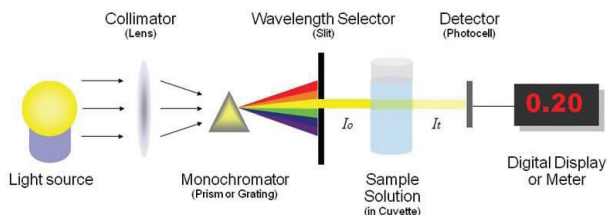
- Also based on Beer-Lambert Law but uses monochromatic light.
- Capable of operating across UV, Visible, and Near-IR regions (200–1000 nm).

#### Types

1. Single-beam spectrophotometer – Measures one sample at a time.
2. Double-beam spectrophotometer – Simultaneously compares sample and reference.

#### Components

1. Light source:
  - UV range: Deuterium lamp
  - Visible range: Tungsten-halogen lamp
2. Monochromator:
  - Prism or diffraction grating used to isolate specific wavelength.
3. Sample holder:
  - Quartz cuvettes for UV; glass/plastic for visible.
4. Detector:
  - Photodiode or photomultiplier tube.
5. Readout system:
  - Digital display or computer interface.



### Working Principle of Spectrophotometer

### Applications

- Quantitative analysis of DNA/RNA (260 nm) and protein (280 nm).
- Determination of enzyme kinetics and reaction rates.
- Pharmaceutical quality control.
- Environmental monitoring (e.g., nitrate, phosphate in water).
- Used in molecular biology, clinical diagnostics, biotechnology, and forensic science.

### Advantages over Colorimeter

- Higher sensitivity and accuracy.
- Measures across broader wavelength range (UV to visible).
- Suitable for colorless compounds absorbing UV light.

### Comparison: Colorimeter Vs Spectrophotometer

| Feature              | Colorimeter          | Spectrophotometer                  |
|----------------------|----------------------|------------------------------------|
| Wavelength Range     | 400–700 nm (visible) | 200–1000 nm (UV-Visible)           |
| Light Source         | Tungsten lamp        | Tungsten (Visible), Deuterium (UV) |
| Wavelength Selection | Filters              | Monochromator (grating/prism)      |
| Sensitivity          | Moderate             | High                               |
| Accuracy             | Lower                | Higher                             |
| Cost                 | Less expensive       | More expensive                     |

### Calibration and Maintenance

- Regular calibration using blank, standard solutions, and certified reference materials.
- Clean cuvettes and ensure proper alignment.
- Periodic lamp replacement and software validation.

### Limitations

- Both techniques depend on:
  - Homogeneous sample solution
  - Proper blanking
  - Linear range of absorbance (usually 0.1–1.0 A.U.)

## ELECTROCARDIOGRAPH (ECG)

### Definition

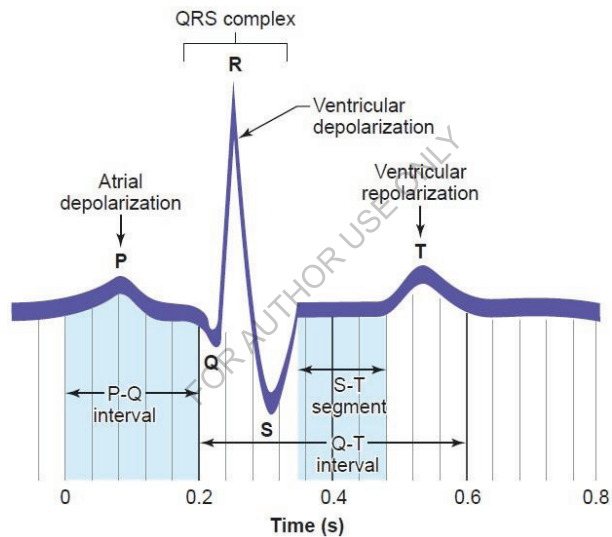
An Electrocardiograph (ECG) is a graphic recording of the electrical activity of the heart as sensed on the body surface. It reflects the conduction system, cardiac rhythm, and electrophysiological integrity of the myocardium.

### Principle

The ECG is based on the fact that the heart generates electrical impulses that depolarize and repolarize the atrial and ventricular muscles. These impulses produce potential differences, which are detected and amplified by surface electrodes placed on the skin.

**Components of ECG**

| Wave/Segment | Description                   | Electrical Event                           | Normal Duration |
|--------------|-------------------------------|--------------------------------------------|-----------------|
| P wave       | First small upward deflection | Atrial depolarization                      | 0.08–0.10 sec   |
| PR interval  | Onset of P to onset of QRS    | AV conduction time                         | 0.12–0.20 sec   |
| QRS complex  | Rapid sharp spike             | Ventricular depolarization                 | ≤ 0.10 sec      |
| ST segment   | Flat segment after QRS        | Ventricular repolarization begins          | -               |
| T wave       | Broad upward wave             | Ventricular repolarization                 | variable        |
| QT interval  | Start of Q to end of T        | Total ventricular activity                 | ≤ 0.44 sec      |
| U wave       | Small wave after T            | Purkinje repolarization (uncertain origin) | Often absent    |



**Electrocardiogram (ECG) Trace Showing P Wave, QRS Complex, T Wave, and Cardiac Cycle Intervals**

**ECG Leads**

Leads are different viewpoints of the heart's electrical activity.

Standard 12-lead ECG:

- a. Limb Leads:
  - Bipolar (Einthoven’s Triangle): I, II, III
  - Unipolar Augmented: aVR, aVL, aVF
- b. Chest Leads (Precordial):
  - V1–V6 (placed on anterior chest wall)

| Lead         | View of Heart                 |
|--------------|-------------------------------|
| I, aVL       | Lateral wall (Left ventricle) |
| II, III, aVF | Inferior wall                 |
| V1-V2        | Septal region                 |
| V3-V4        | Anterior wall                 |
| V5-V6        | Lateral wall                  |

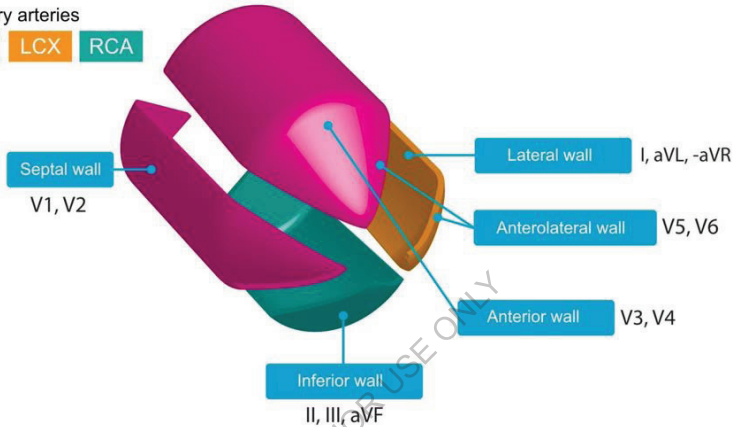
The four walls of the left ventricle and the ECG leads viewing these walls

Coronary arteries

LAD

LCX

RCA



### Anatomical Walls of the Left Ventricle with Corresponding ECG Leads and Coronary Artery Supply (LAD, LCX, RCA)

#### Interpretation of Electrocardiograph

##### Interpretation Steps

- Rate
  - Count R-R intervals.
  - Normal rate: 60–100 bpm (sinus rhythm).
- Rhythm
  - Is it regular?
  - Are P waves present before every QRS?
- Axis Deviation
  - Look at Leads I and aVF:
    - Normal Axis: Lead I & aVF positive
    - Left Axis Deviation (LAD): I positive, aVF negative
    - Right Axis Deviation (RAD): I negative, aVF positive.
- Waveform Analysis
  - P wave morphology (RA/LA enlargement)
  - QRS complex (amplitude, duration, bundle branch blocks)
  - ST segment (elevation = MI, depression = ischemia)

- T wave (inversion, hyperacute)
- QT interval (prolongation indicates risk of arrhythmias)

#### 5. Pathological Q Wave

- 0.04 sec duration and >25% of R wave in same lead
- Indicates old myocardial infarction

#### 6. Common Abnormal Patterns

| Condition                  | ECG Finding                                                            |
|----------------------------|------------------------------------------------------------------------|
| Myocardial infarction (MI) | ST elevation/depression, Q waves                                       |
| Arrhythmias                | Irregular rhythm, absent P waves, abnormal QRS                         |
| Atrial fibrillation        | No P waves, irregularly irregular rhythm                               |
| Heart block                | Prolonged PR (1°), dropped QRS (2°), dissociation (3°)                 |
| Ventricular hypertrophy    | Increased QRS amplitude, strain patterns                               |
| Electrolyte imbalance      | Tall T waves ( $\uparrow K^+$ ), prolonged QT ( $\downarrow Ca^{2+}$ ) |
| Pericarditis               | Diffuse ST elevation, PR depression                                    |

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